Storyline

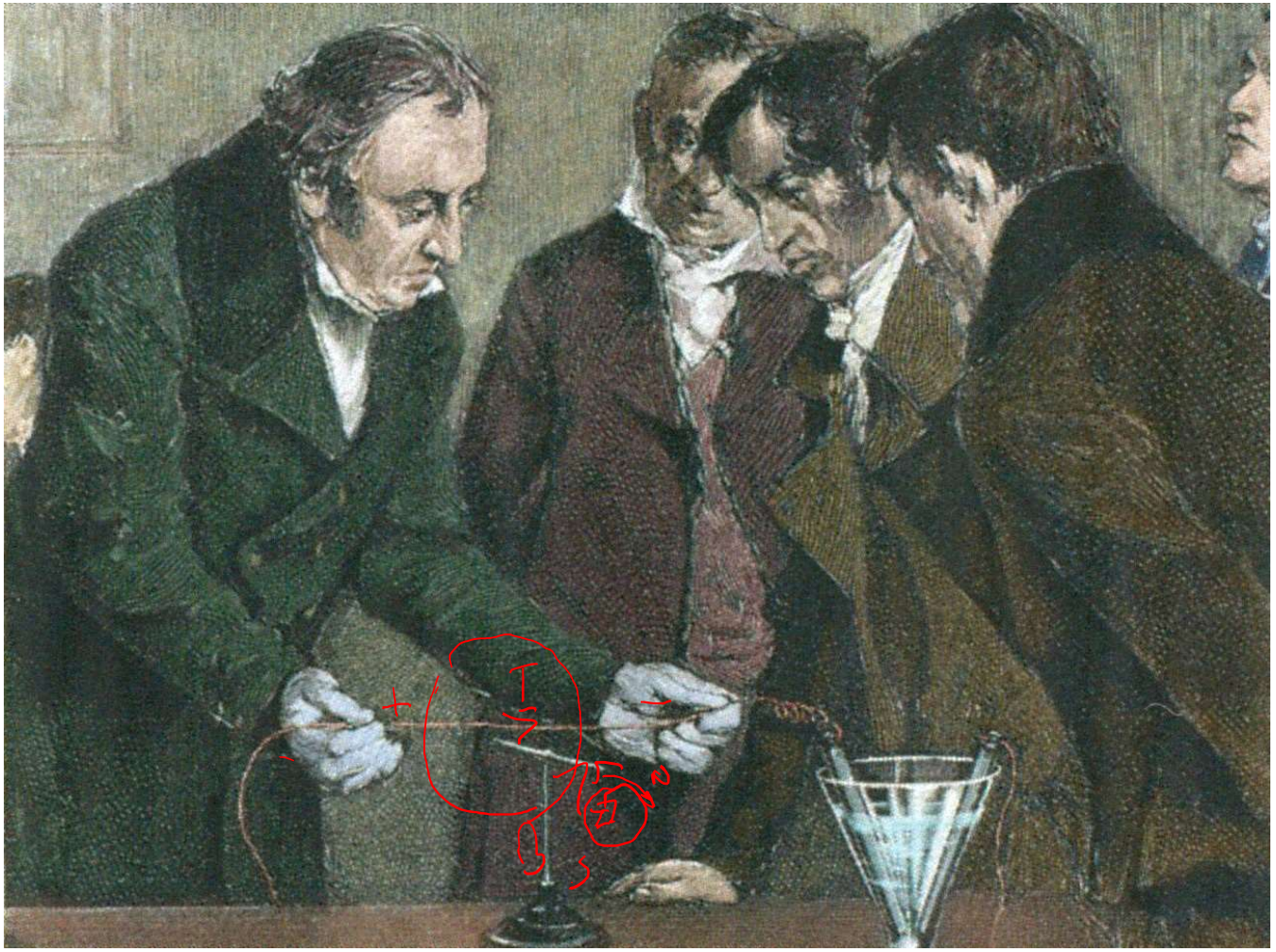
Chapter 29: Sources of the Magnetic Field



Physics for Scientists and Engineers, 10e
Raymond A. Serway
John W. Jewett, Jr.

BROOKS/COLE
CENGAGE Learning

Relationship between Magnetism and Electricity



The Biot–Savart Law

magnetic field due to current through length element

direction of $\vec{B}$

$d\vec{s}$ = length of segment = Length Element

$$d\vec{\mathbf{B}} = \frac{\mu_0}{4\pi}\frac{I d\vec{\mathbf{s}} \times \hat{\mathbf{r}}}{r^2}$$

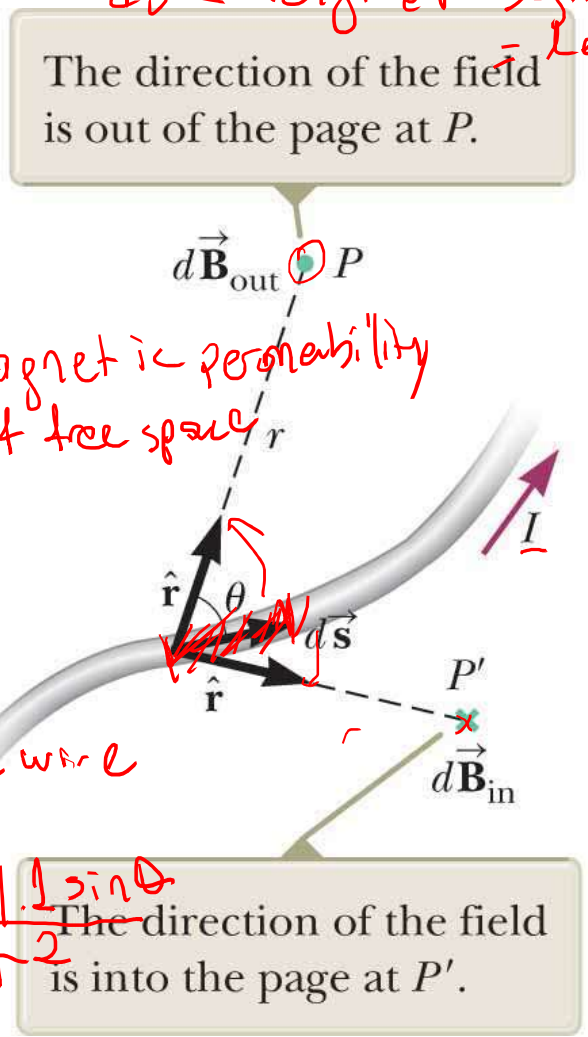
$$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{T \cdot m/A}$$ → magnetic permeability of free space

$|\hat{r}| = 1$

$$\vec{\mathbf{E}} = k_e \frac{q}{r^2}\hat{\mathbf{r}}$$

Total $\vec{B}$ due to I through the wire

$$\vec{\mathbf{B}} = \frac{\mu_0 I}{4\pi}\int \frac{d\vec{\mathbf{s}} \times \hat{\mathbf{r}}}{r^2} = \frac{\mu_0 I}{4\pi}\int \frac{|ds| \cdot 1 \sin\theta}{r^2}$$



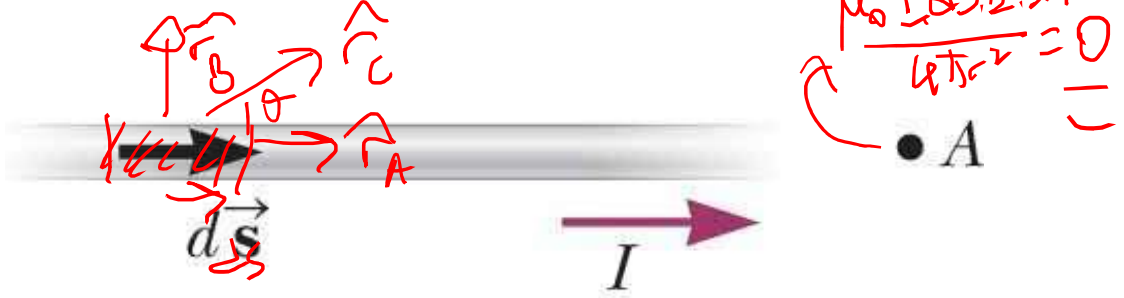
Quick Quiz 29.1

Consider the magnetic field due to the current in the wire shown in the figure. Rank the points A, B, and C in terms of magnitude of the magnetic field that is due to the current in just the length element $d\vec{\mathbf{s}}$ shown from greatest to least.

$$d\vec{B} = \frac{\mu_0 I}{4\pi}\frac{d\vec{s}\times\hat{r}}{r^2}$$

$$\frac{\mu_0 I\, ds \cdot 1 \sin 90}{4\pi r^2}$$

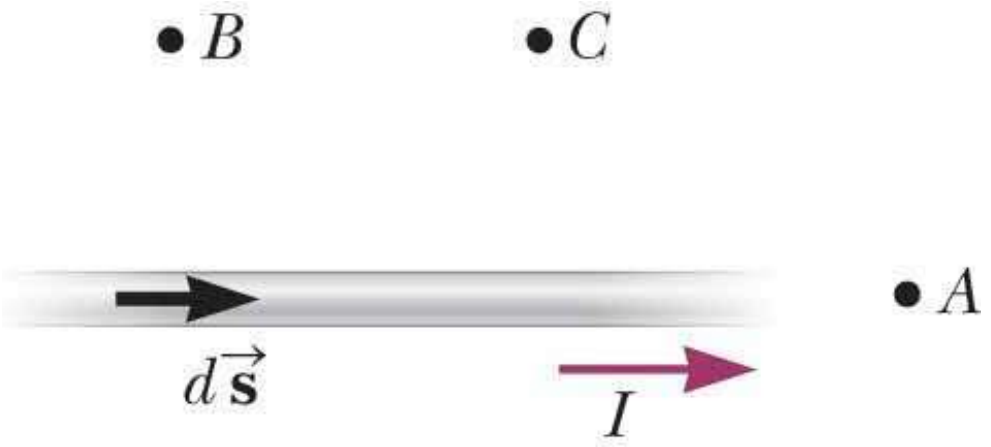
$$\frac{\mu_0 I\, ds \cdot 1 \sin\alpha}{4\pi r^2}$$



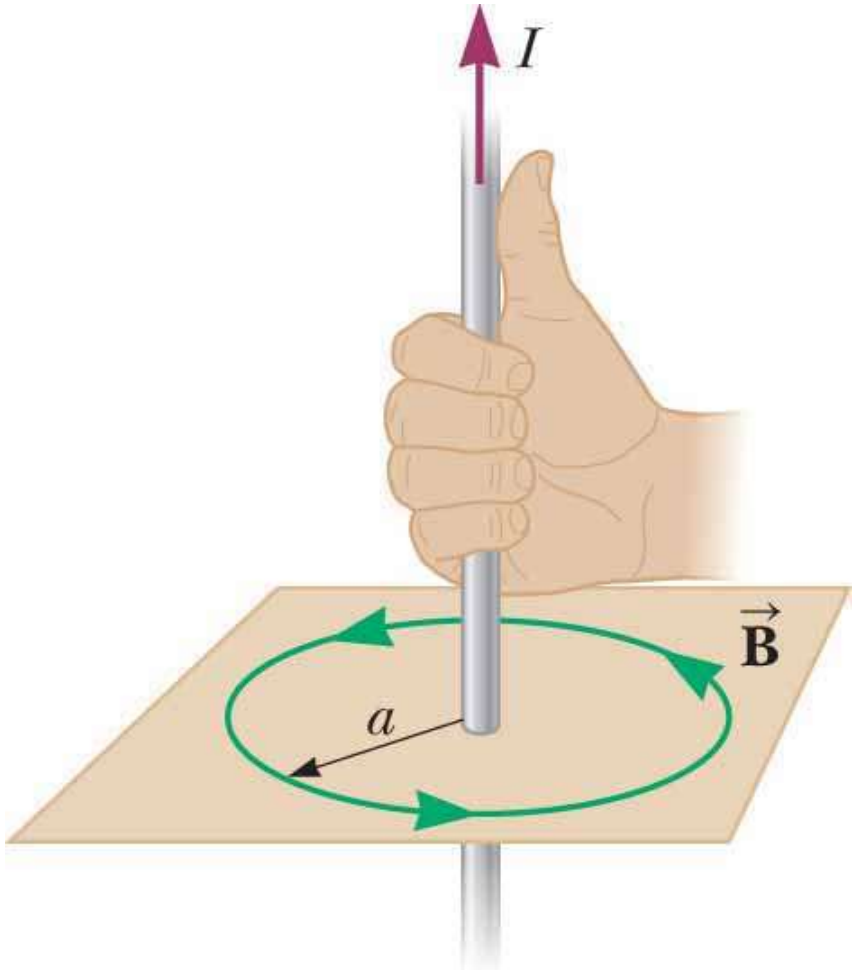
Quick Quiz 29.1

Consider the magnetic field due to the current in the wire shown in the figure. Rank the points A, B, and C in terms of magnitude of the magnetic field that is due to the current in just the length element $d\vec{\mathbf{s}}$ shown from greatest to least.

B* > *C* > *A

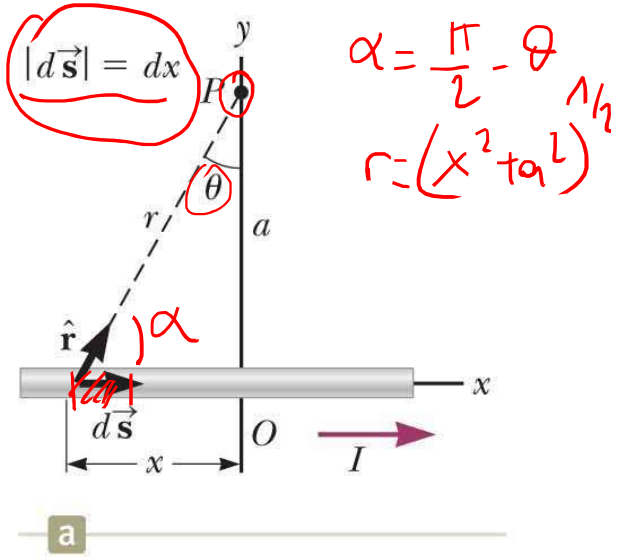


Magnetic Fields Due to Currents



Example 29.1: Magnetic Field Surrounding a Thin, Straight Conductor

Consider a thin, straight wire of finite length carrying a constant current I and placed along the x axis as shown in the figures. Determine the magnitude and direction of the magnetic field at point P due to this current.

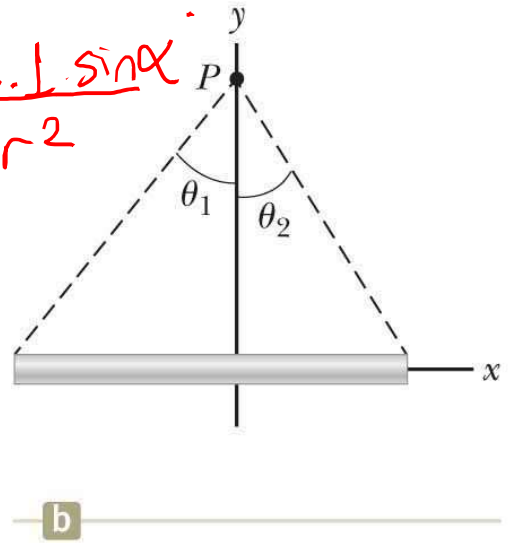


$\alpha = \frac{\pi}{2} - \theta$

$r = (x^2 + a^2)^{1/2}$

$$dB = \frac{\mu_0}{4\pi}\frac{I\,d\vec{s}\times\hat{r}}{r^2} = \frac{\mu_0 I}{4\pi}\frac{dx\,\sin\alpha}{r^2}$$

$$dB = \frac{\mu_0 I\,dx}{r^2}\sin\left(\frac{\pi}{2} - \theta\right)$$



Example 29.1: Magnetic Field Surrounding a Thin, Straight Conductor

$$d\vec{\mathbf{s}} \times \hat{\mathbf{r}} = \left| d\vec{\mathbf{s}} \times \hat{\mathbf{r}} \right| \hat{\mathbf{k}} = \left[dx \sin\left(\frac{\pi}{2} - \theta \right) \right] \hat{\mathbf{k}}$$

$$= \left(dx \cos\theta \right) \hat{\mathbf{k}}$$

$$d\vec{\mathbf{B}} = \left(dB \right) \hat{\mathbf{k}} = \frac{\mu_0 I}{4\pi} \frac{dx \cos\theta}{r^2} \hat{\mathbf{k}}$$

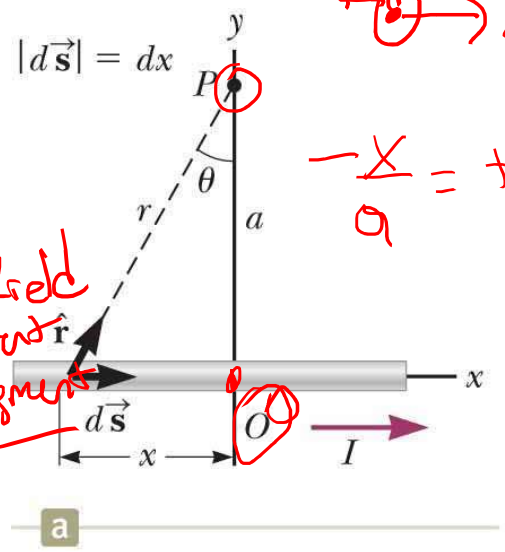
magnetic field due to current through segment

$$r = a / \cos\theta \qquad x = -a \tan\theta$$

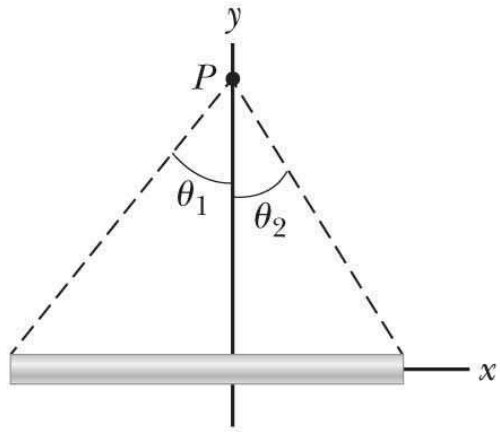
$$dx = -a \sec^2\theta \, d\theta = -\frac{a \, d\theta}{\cos^2\theta}$$

$$dB = -\frac{\mu_0 I}{4\pi} \left(\frac{a \, d\theta}{\cos^2\theta} \right) \left(\frac{\cos^2\theta}{a^2} \right) \cos\theta$$

$$= -\frac{\mu_0 I}{4\pi a} \cos\theta \, d\theta$$



a



b

Example 29.1: Magnetic Field Surrounding a Thin, Straight Conductor

$$dB = -\frac{\mu_0 I}{4\pi}\left(\frac{a\,d\theta}{\cos^2\theta}\right)\left(\frac{\cos^2\theta}{a^2}\right)\cos\theta = -\frac{\mu_0 I}{4\pi a}\cos\theta\,d\theta$$

Total magnetic over all segments

$$B = -\frac{\mu_0 I}{4\pi a}\int_{\theta_1}^{\theta_2}\cos\theta\,d\theta = \boxed{\frac{\mu_0 I}{4\pi a}\left(\sin\theta_1 - \sin\theta_2\right)}$$

→ finite length

Case for infinitely long wire

$$\left(\sin\theta_1 - \sin\theta_2\right) = \left[\sin\pi/2 - \sin\left(-\pi/2\right)\right] = 2$$

$\theta_1 = \pi/2$ $\quad \theta_2 = -\pi/2$ $\quad \tan\theta = \frac{x}{a}$

Magnetic field due to current through infinite wire.

$$B = \frac{\mu_0 I}{2\pi a} \rightarrow \infty$$

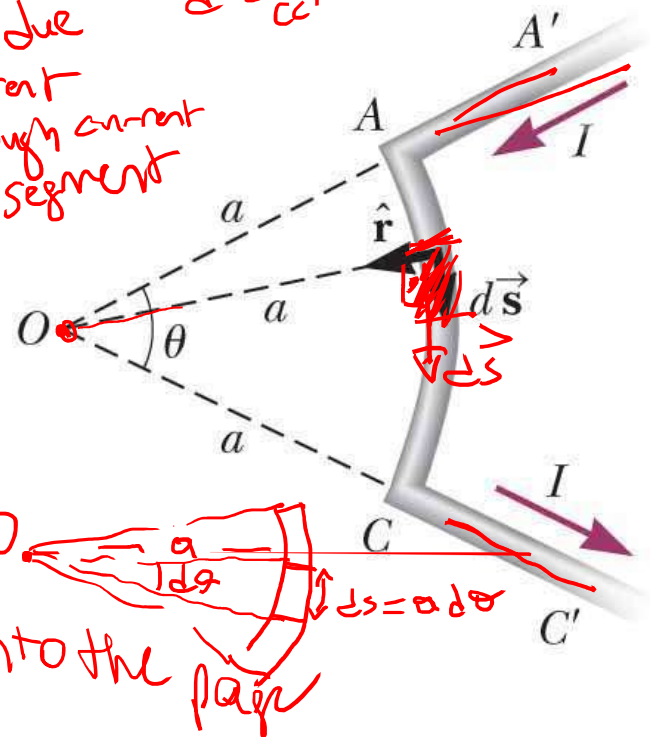
Example 29.2: Magnetic Field Due to a Curved Wire Segment

Calculate the magnetic field at point *O* for the current-carrying wire segment shown in the figure. The wire consists of two straight portions and a circular arc of radius a, which subtends an angle θ.

$$dB = \frac{\mu_0}{4\pi}\frac{I\,ds}{a^2}$$

$$B = \frac{\mu_0 I}{4\pi a^2}\int ds = \frac{\mu_0 I}{4\pi a^2}s$$

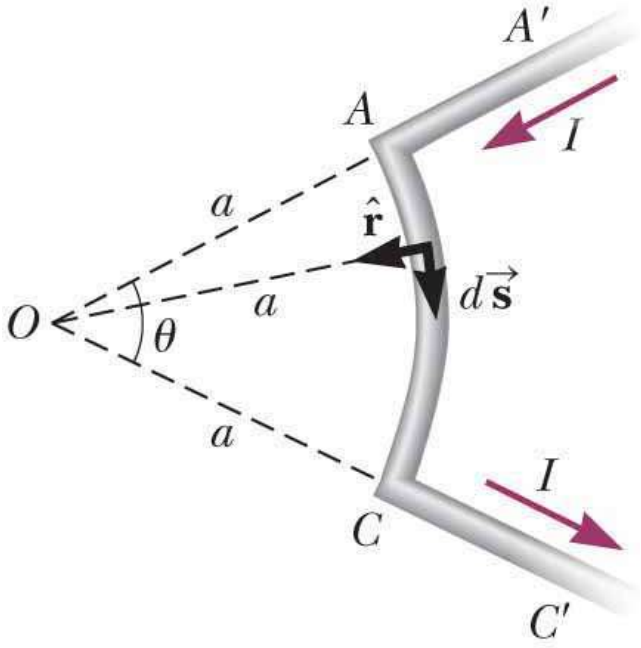
$$B = \frac{\mu_0 I}{4\pi a^2}(a\theta) = \frac{\mu_0 I}{4\pi a}\theta$$



Example 29.2: Magnetic Field Due to a Curved Wire Segment

What if you were asked to find the magnetic field at the center of a circular wire loop of radius R that carries a current I? Can this question be answered at this point in our understanding of the source of magnetic fields?

$$B = \frac{\mu_0 I}{4\pi a} 2\pi = \frac{\mu_0 I}{2a}$$



Example 29.3: Magnetic Field on the Axis of a Circular Current Loop

Consider a circular wire loop of radius a located in the *yz* plane and carrying a steady current *I* as in the figure. Calculate the magnetic field at an axial point *P* a distance *x* from the center of the loop.

$$d\vec{B} = \frac{\mu_0 I}{4\pi}\frac{d\vec{s}\times\hat{r}}{r^2} = \frac{\mu_0 I}{4\pi}\frac{ds\cdot 1\cdot\sin 90^\circ}{r^2}$$

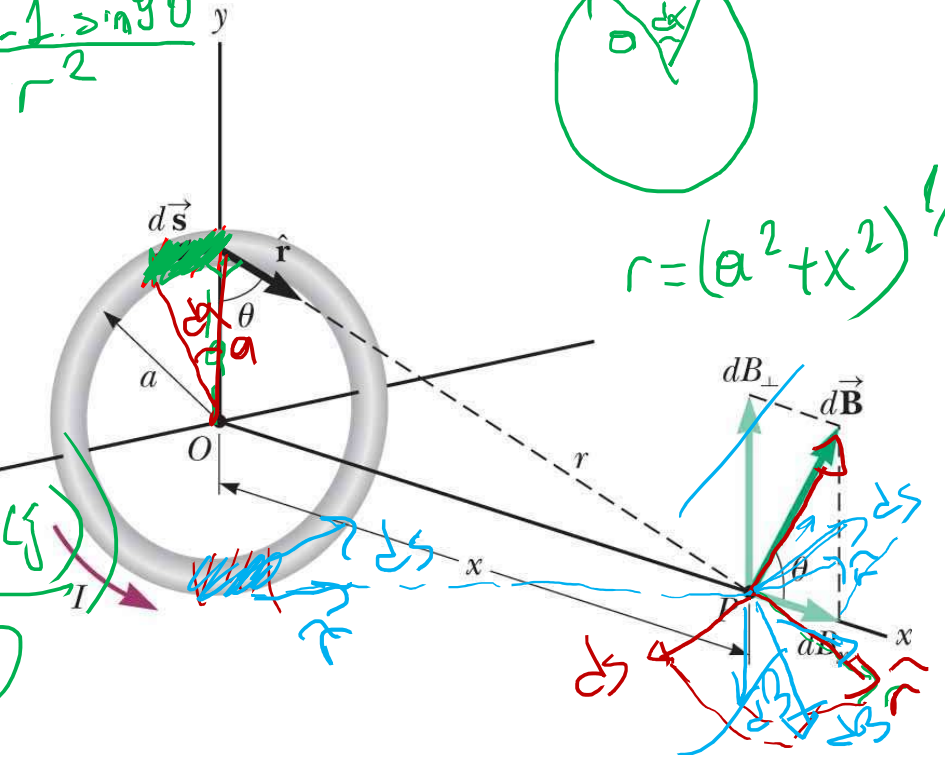
$$d\vec{B} = \frac{\mu_0 I\, ds}{4\pi r^2}$$

$$\int dB = \int \frac{\mu_0 I\, ds}{4\pi r^2}(\cos\theta\,\hat{i} + \sin\theta\,\hat{j})$$

$\neq 0$ $\quad = 0$

$ds = a\,d\alpha$

$r = (a^2 + x^2)^{1/2}$

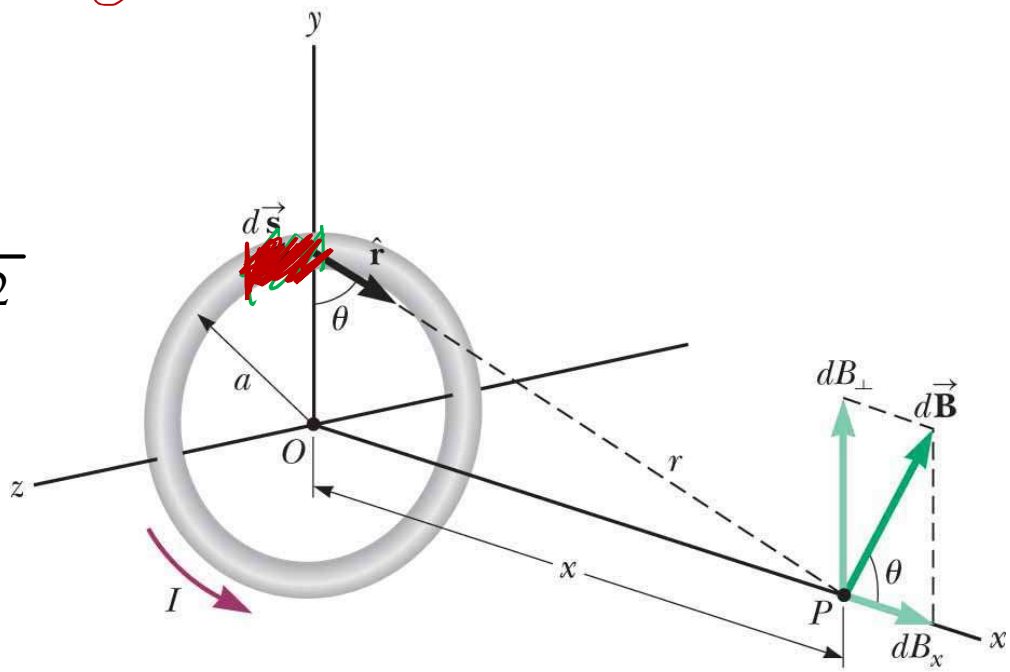


Example 29.3: Magnetic Field on the Axis of a Circular Current Loop

$$dB = \frac{\mu_0 I}{4\pi}\left|\frac{d\vec{\mathbf{s}}\times\hat{\mathbf{r}}}{r^2}\right| = \frac{\mu_0 I}{4\pi}\frac{ds}{\left(a^2+x^2\right)}$$

$$dB_x = \frac{\mu_0 I}{4\pi}\frac{ds}{\left(a^2+x^2\right)}\cos\theta$$

$$\cos\theta = \frac{a}{\left(a^2+x^2\right)^{1/2}}$$

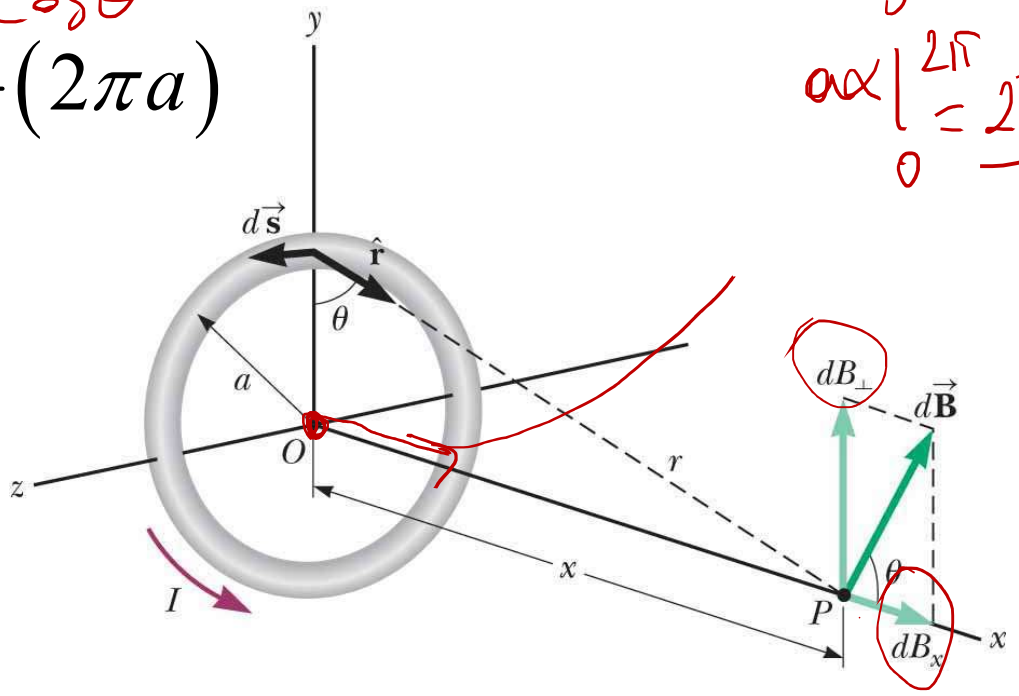


Example 29.3: Magnetic Field on the Axis of a Circular Current Loop

$$B_x = \frac{\mu_0 I}{4\pi}\oint \frac{ds}{a^2+x^2}\left[\frac{a}{\left(a^2+x^2\right)^{1/2}}\right] = \frac{\mu_0 I}{4\pi}\frac{a}{\left(a^2+x^2\right)^{3/2}}\oint ds$$

$$B_x = \frac{\mu_0 I}{4\pi}\frac{a}{\left(a^2+x^2\right)^{3/2}}(2\pi a)$$

$$= \boxed{\frac{\mu_0 I a^2}{2\left(a^2+x^2\right)^{3/2}}}$$



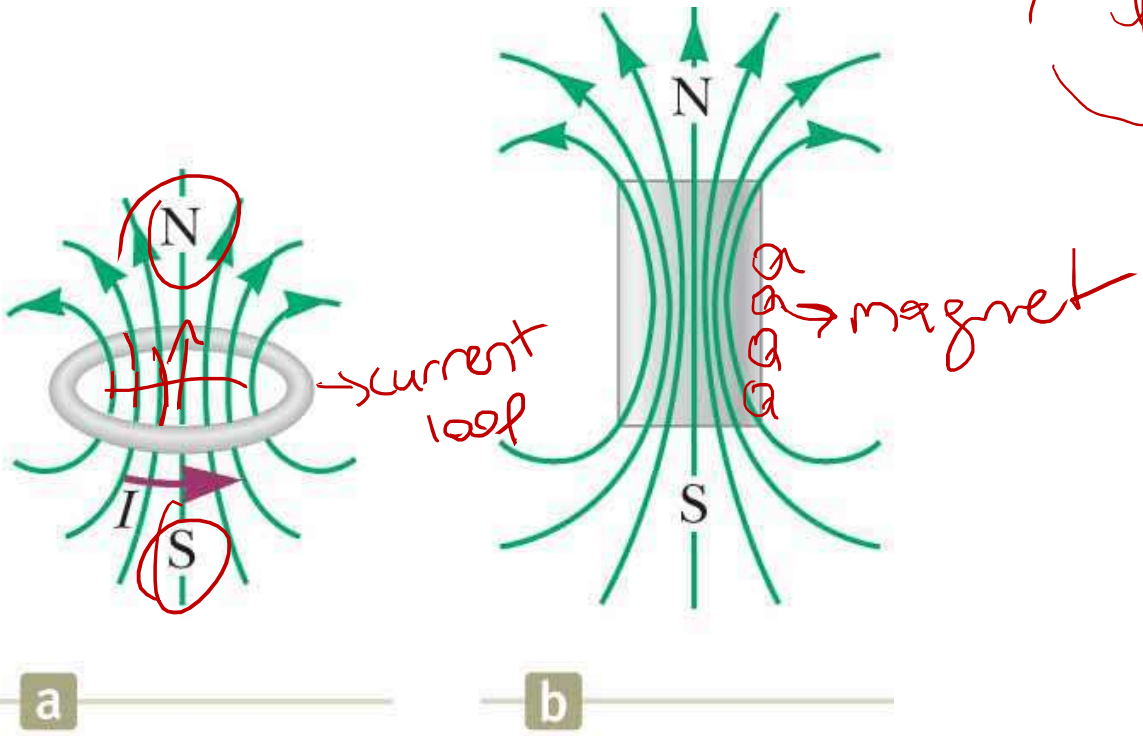
Example 29.3: Magnetic Field on the Axis of a Circular Current Loop

$$B = \frac{\mu_0 I a^2}{2(a^2 + x^2)^{3/2}} = \frac{\mu_0 I}{2a}$$

$x = 0$

$$B = \frac{\mu_0 I}{2a} \quad (\text{at center of loop})$$



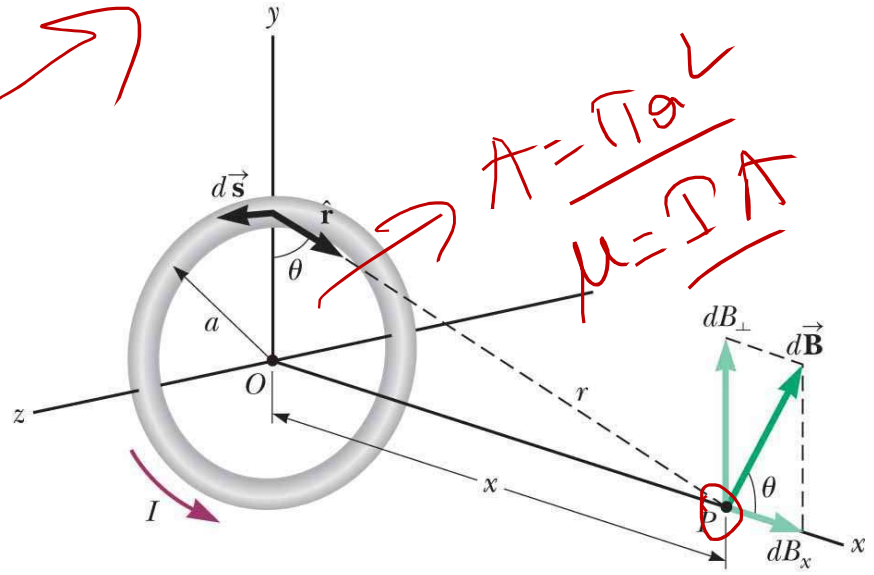


Example 29.3: Magnetic Field on the Axis of a Circular Current Loop

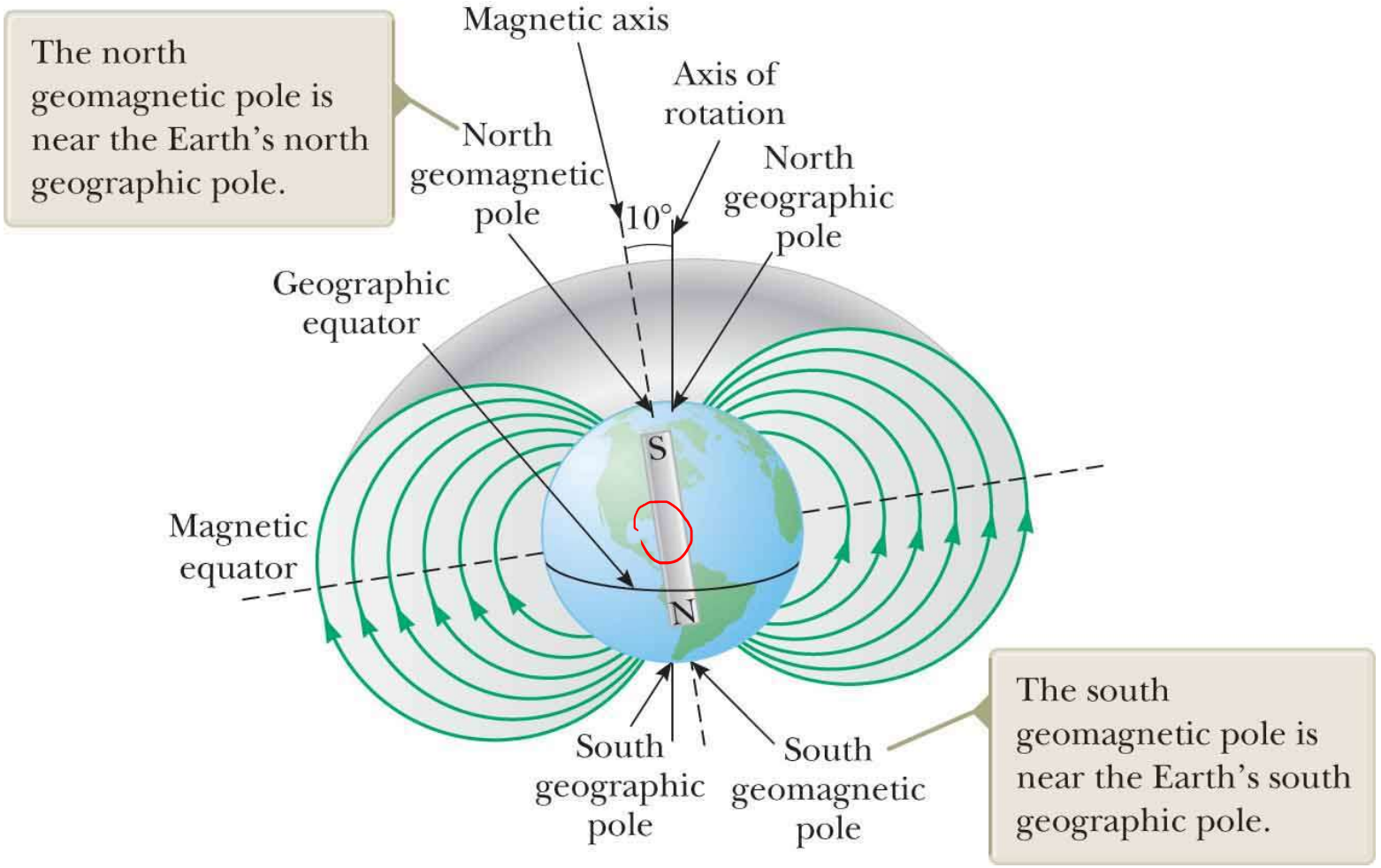
What if we consider points on the x axis very far from the loop? How does the magnetic field behave at these distant points?

$$B_x = \frac{\mu_0 I a^2}{2\left(a^2 + x^2\right)^{3/2}} \Rightarrow B \approx \frac{\mu_0 I a^2}{2x^3} \left(\text{for } x >> a\right)$$

$$B \approx \frac{\mu_0}{2\pi}\frac{\mu}{x^3}$$



Source of Earth's Magnetic Field



The Magnetic Force Between Two Parallel Conductors

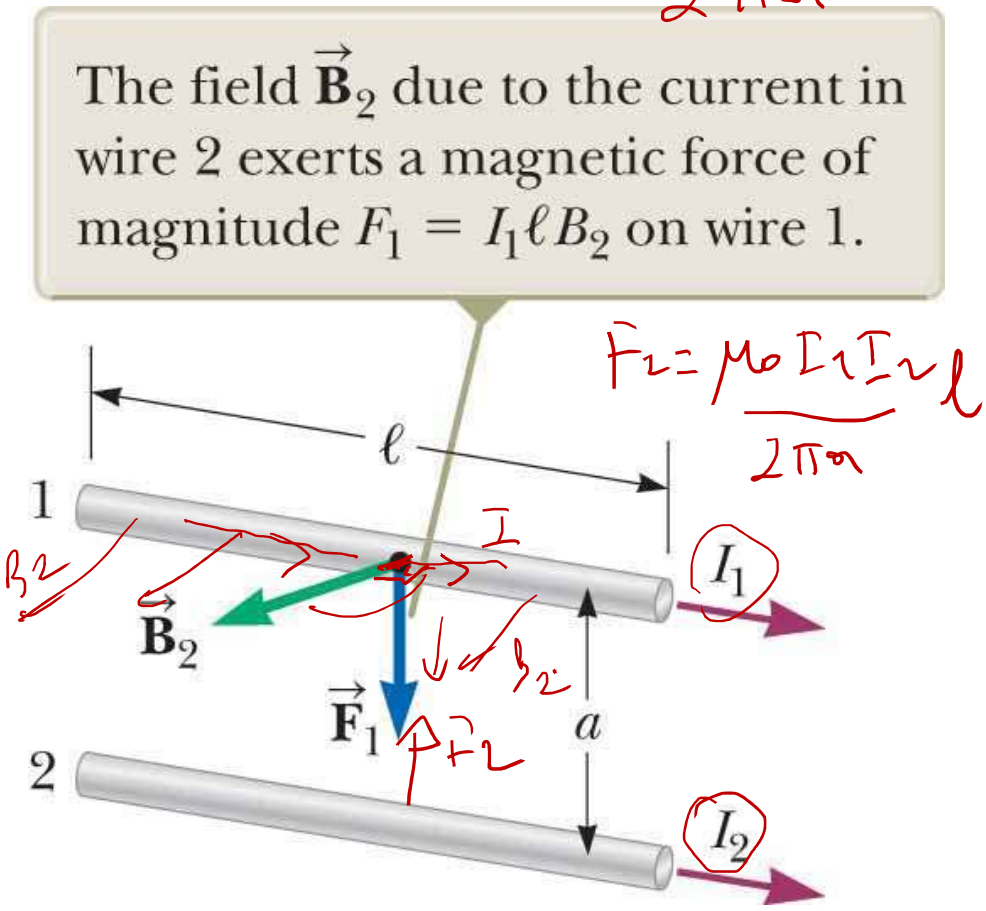
$$\vec{\mathbf{F}}_1 = I_1 \vec{\ell} \times \vec{\mathbf{B}}_2 \qquad F_1 = I_1 \ell B_2$$

$$F_1 = I_1 \ell B_2$$

$$= I_1 \ell \left(\frac{\mu_0 I_2}{2\pi a} \right)$$

$$F_1 = \frac{\mu_0 I_1 I_2}{2\pi a} \ell$$

$$\frac{F_B}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi a}$$



The Magnetic Force Between Two Parallel Conductors

When the magnitude of the force per unit length between two long, parallel wires that carry identical currents and are separated by 1 m is 2×10^{-7} N/m, the current in each wire is defined to be 1 A.

$$\frac{F_B}{\ell} = \frac{\mu_0 I_1 I_2}{2\pi a} \qquad F_1 = \frac{\mu_0 I_1 I_2}{2\pi a}\ell$$

Quick Quiz 29.2

A loose spiral spring carrying no current is hung from a ceiling. When a switch is thrown so that a current exists in the spring, do the coils

(a) move closer together,

(b) move farther apart, or

(c) not move at all?



Quick Quiz 29.2

A loose spiral spring carrying no current is hung from a ceiling. When a switch is thrown so that a current exists in the spring, do the coils

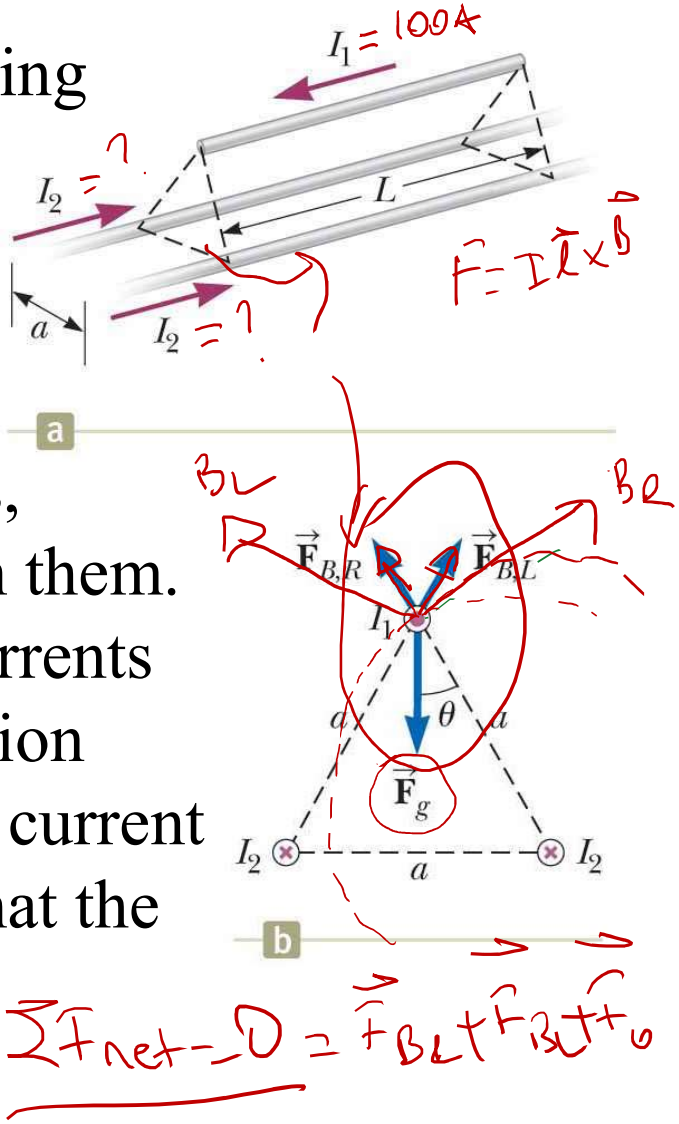
(a) move closer together,

(b) move farther apart, or

(c) not move at all?

Example 29.4: Suspending a Wire

Two infinitely long, parallel wires are lying on the ground a distance a = 1.00 cm apart as shown in the top figure. A third wire, of length L = 10.0 m and mass 400 g, carries a current of I_1 = 100 A and is levitated above the first two wires, at a horizontal position midway between them. The infinitely long wires carry equal currents I_2 in the same direction, but in the direction opposite that in the levitated wire. What current must the infinitely long wires carry so that the three wires form an equilateral triangle?



Example 29.4: Suspending a Wire

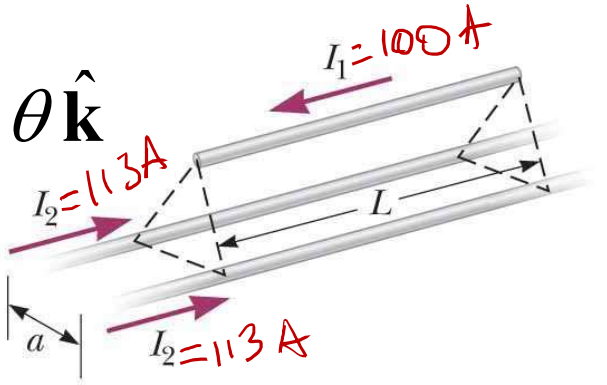
$$F_B = 2\left(\frac{\mu_0 I_1 I_2}{2\pi a}\ell\right)\cos\theta\,\hat{\mathbf{k}} = \frac{\mu_0 I_1 I_2}{2\pi a}\ell\cos\theta\,\hat{\mathbf{k}}$$

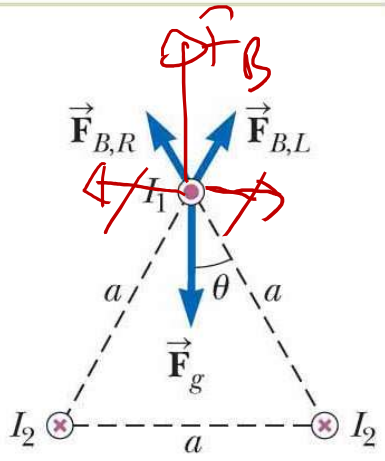
$$\vec{\mathbf{F}}_g = -mg\hat{\mathbf{k}}$$

$$\sum\vec{\mathbf{F}} = \vec{\mathbf{F}}_B + \vec{\mathbf{F}}_g$$

$$= \frac{\mu_0 I_1 I_2}{\pi a}\ell\cos\theta\,\hat{\mathbf{k}} - mg\hat{\mathbf{k}} = 0$$

$$I_2 = \frac{mg\pi a}{\mu_0 I_1 \ell\cos\theta}$$





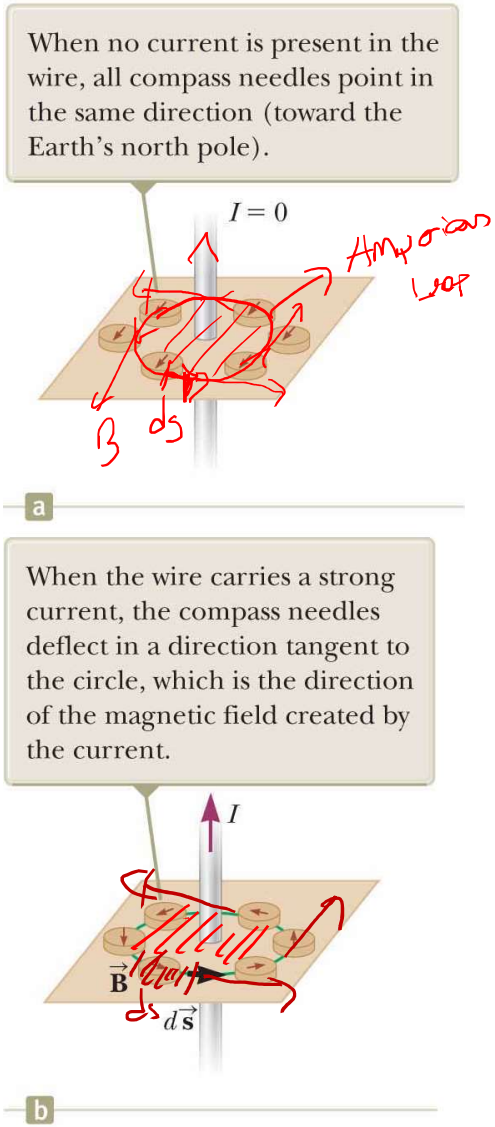
$$I_2 = \frac{(0.400\text{ kg})(9.80\text{ m/s}^2)\pi(0.0100\text{ m})}{(4\pi\times10^{-7}\text{ T}\cdot\text{m/A})(100\text{ A})(10.0\text{ m})\cos 30.0^\circ} = 113\text{ A}$$

Ampère's Law

$$\oint \vec{B} \cdot d\vec{s} = B \oint ds = \frac{\mu_0 I}{2\pi r}(2\pi r) = \mu_0 I$$

The line integral of $\vec{B} \cdot d\vec{s}$ around any closed path equals $\mu_0 I$, where I is the total steady current passing through any surface bounded by the closed path:

$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$$



Quick Quiz 29.3

Rank the magnitudes of $\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}}$ for the closed paths a through d in the figure from greatest to least.

$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$

for a: $(1+5-2)\mu_0$

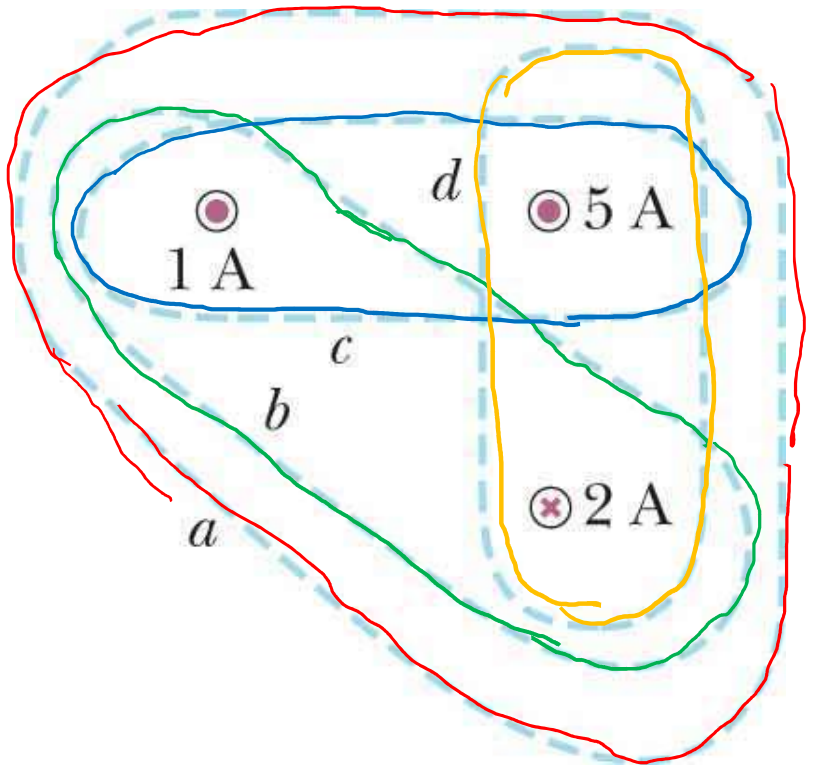
$= 4\mu_0$

b: $(1-2)\mu_0 = -\mu_0$

c $(1+5)\mu_0 = 6\mu_0$

d $(5-2)\mu_0 = 3\mu_0$

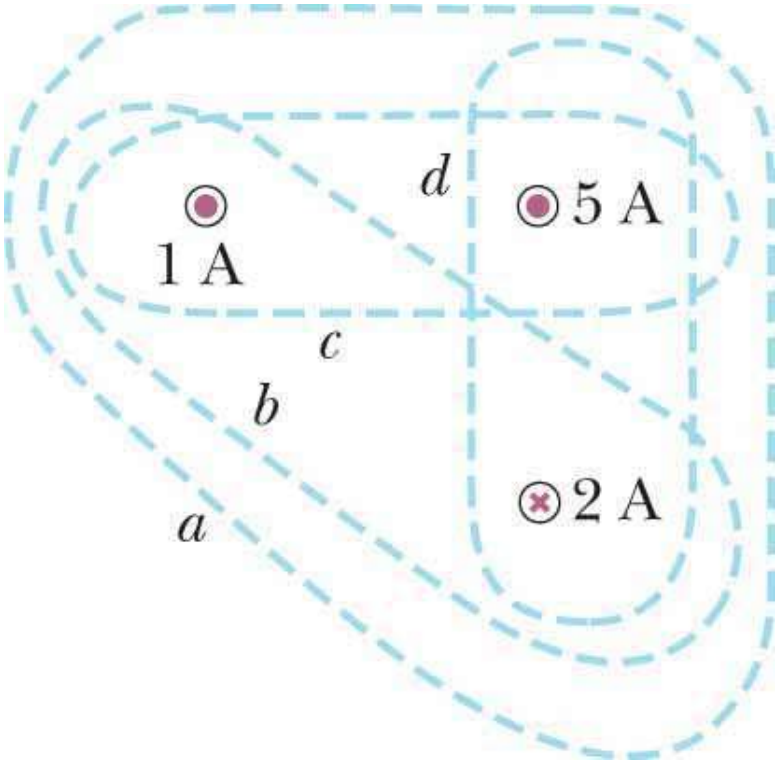
c, a, d, b



Quick Quiz 29.3

Rank the magnitudes of $\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}}$ for the closed paths a through d in the figure from greatest to least.

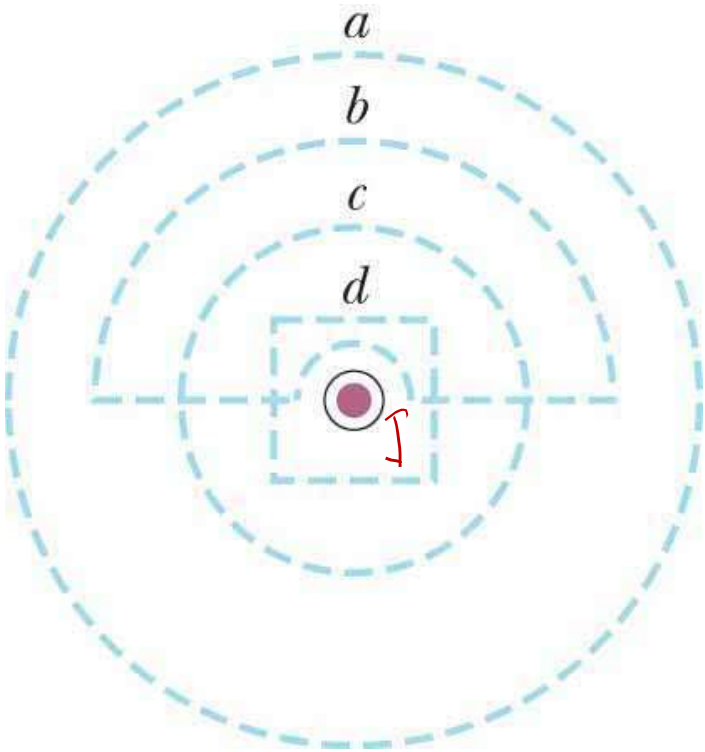
c > a > d > b



Quick Quiz 29.4

Rank the magnitudes of $\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}}$ for the closed paths a through d in the figure from greatest to least.

a → $\mu_0 I$
b → 0
c → $\mu_0 I$
d → $\mu_0 I$

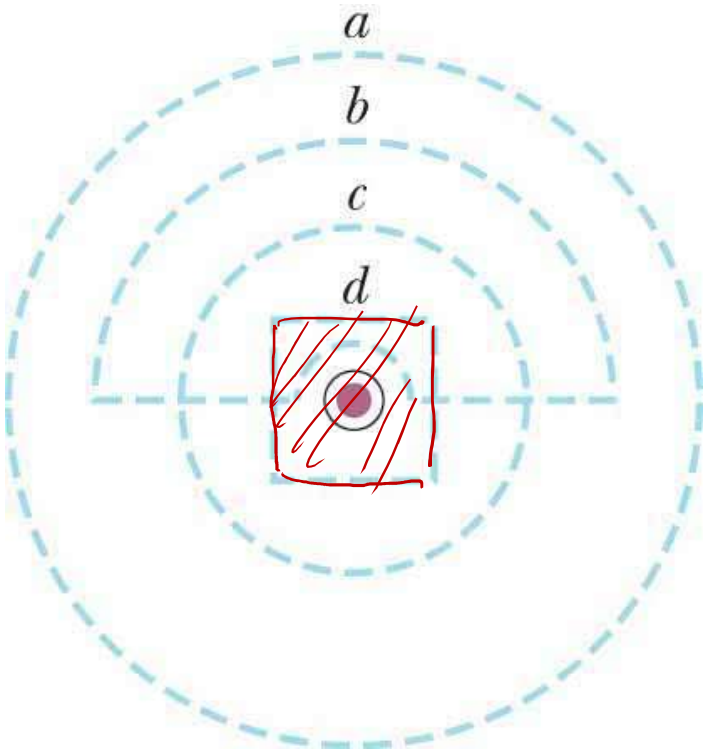


Quick Quiz 29.4

Rank the magnitudes of $\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}}$ for the closed paths a through d in the figure from greatest to least.

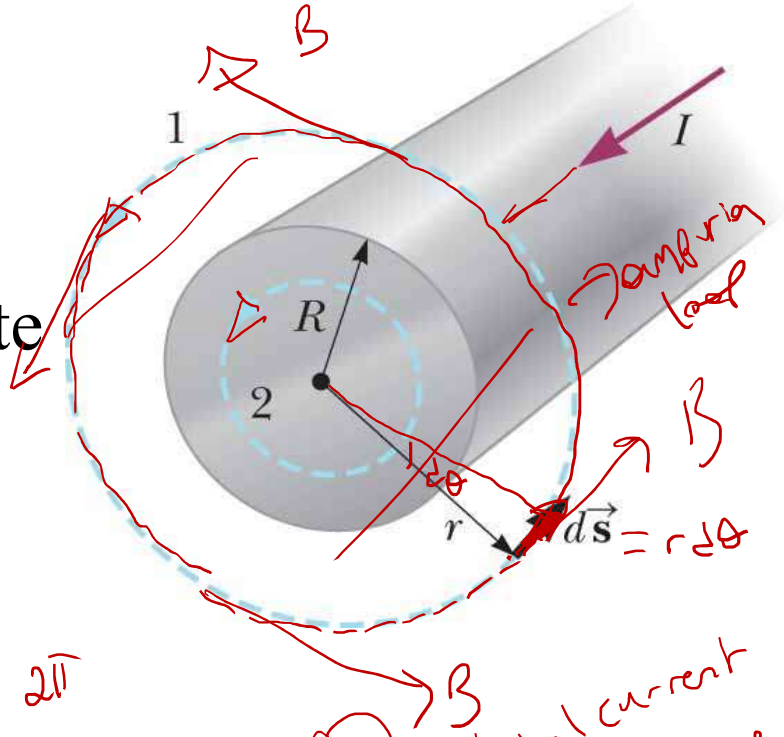
$\mu_0 I$

$$a = c = d > b = 0$$



Example 29.5: The Magnetic Field Created by a Long Current-Carrying Wire

A long, straight wire of radius R carries a steady current *I* that is uniformly distributed through the cross section of the wire. Calculate the magnetic field a distance *r* from the center of the wire in the regions $r \geq R$ and $r < R$.



$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$

$\int |B|(ds)\cos 0 = \mu_0 I_{enc}$

$B \int ds = \mu_0 I_{enc} \rightarrow$

$B \int_0^{2\pi} r\,d\theta = \mu_0 I \rightarrow$ total current passing through the Amperian Loop

$B \cdot 2\pi r = \mu_0 I$

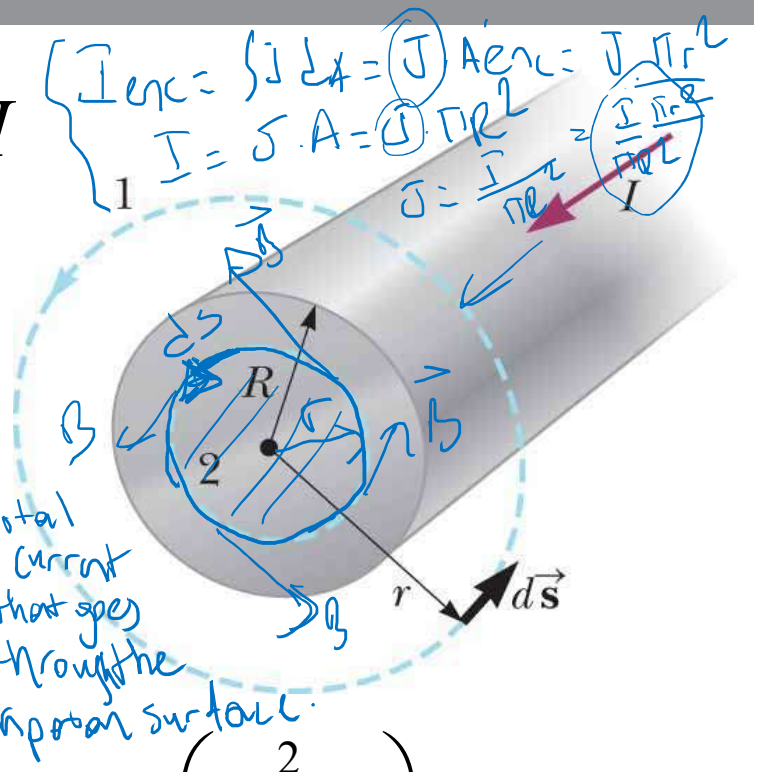
$B = \frac{\mu_0 I}{2\pi r}$ $r \geq R$

Example 29.5: The Magnetic Field Created by a Long Current-Carrying Wire

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = B \oint ds = B(2\pi r) = \mu_0 I$$

$$B = \boxed{\frac{\mu_0 I}{2\pi r}} \quad (\text{for } r \geq R)$$

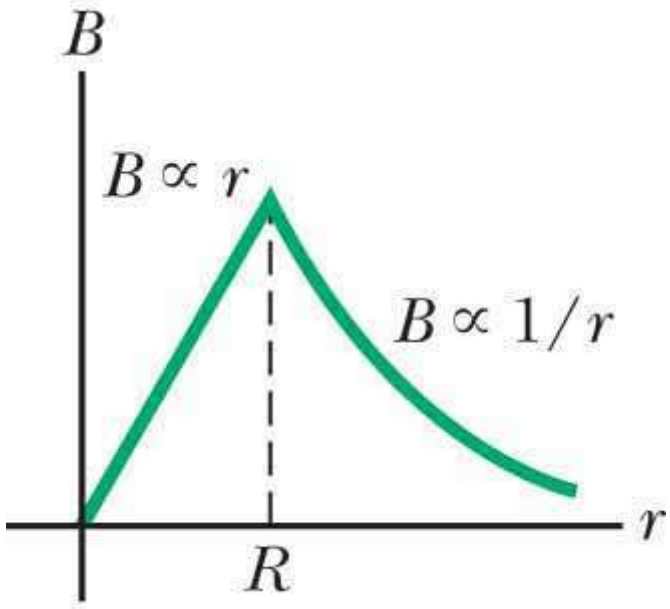
$$\frac{I'_{enc}}{I} = \frac{JA'}{JA} = \frac{\pi r^2}{\pi R^2} \quad \Rightarrow I'_{enc} = \frac{r^2}{R^2} I$$

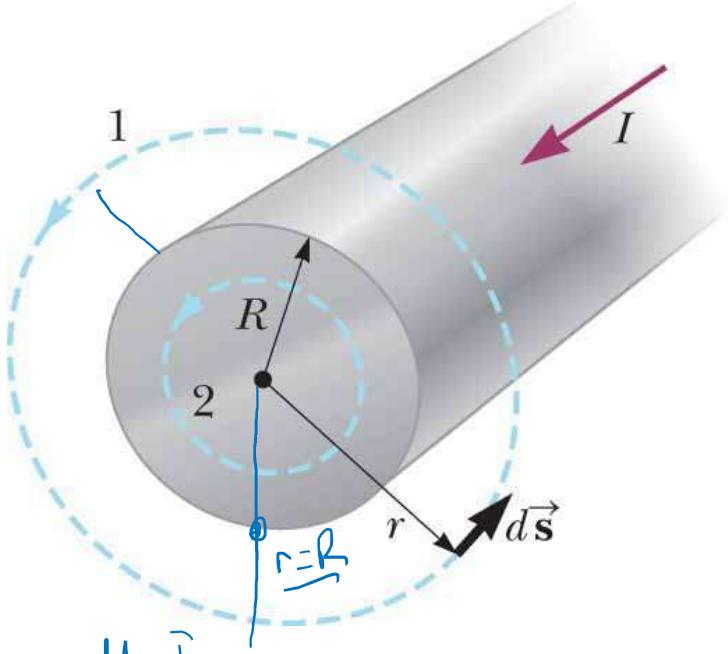


$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = B(2\pi r) = \mu_0 I'_{enc} = \mu_0 \left(\frac{r^2}{R^2} I \right)$$

$$B = \boxed{\left(\frac{\mu_0 I}{2\pi R^2} \right) r} \quad (\text{for } r < R)$$

Example 29.5: The Magnetic Field Created by a Long Current-Carrying Wire





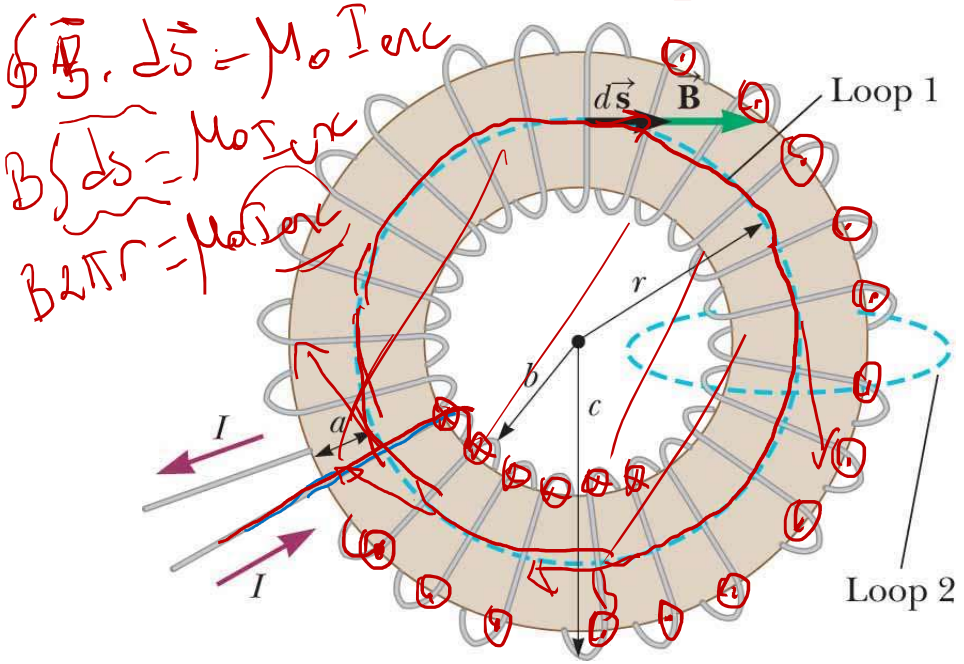
$$B = \frac{\mu_0 I}{2\pi r} \quad (\text{for } r \geq R)$$

$$B = \left(\frac{\mu_0 I}{2\pi R^2}\right) r \quad (\text{for } r < R)$$

$$B = \frac{\mu_0 I}{2\pi R}$$

Example 29.6: The Magnetic Field Created by a Toroid

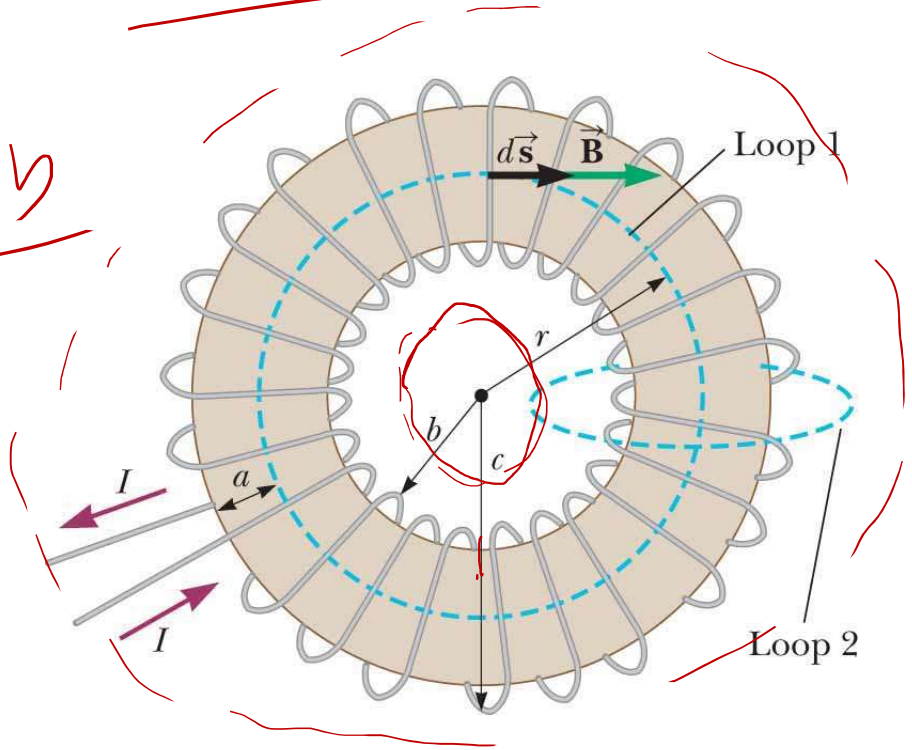
A device called a toroid is often used to create an almost uniform magnetic field in some enclosed area. The device consists of a conducting wire wrapped around a ring (a torus), which is shaped like a doughnut, made of a nonconducting material. For a toroid having *N* closely spaced turns of wire, calculate the magnetic field in the region occupied by the torus, a distance *r* from the center.



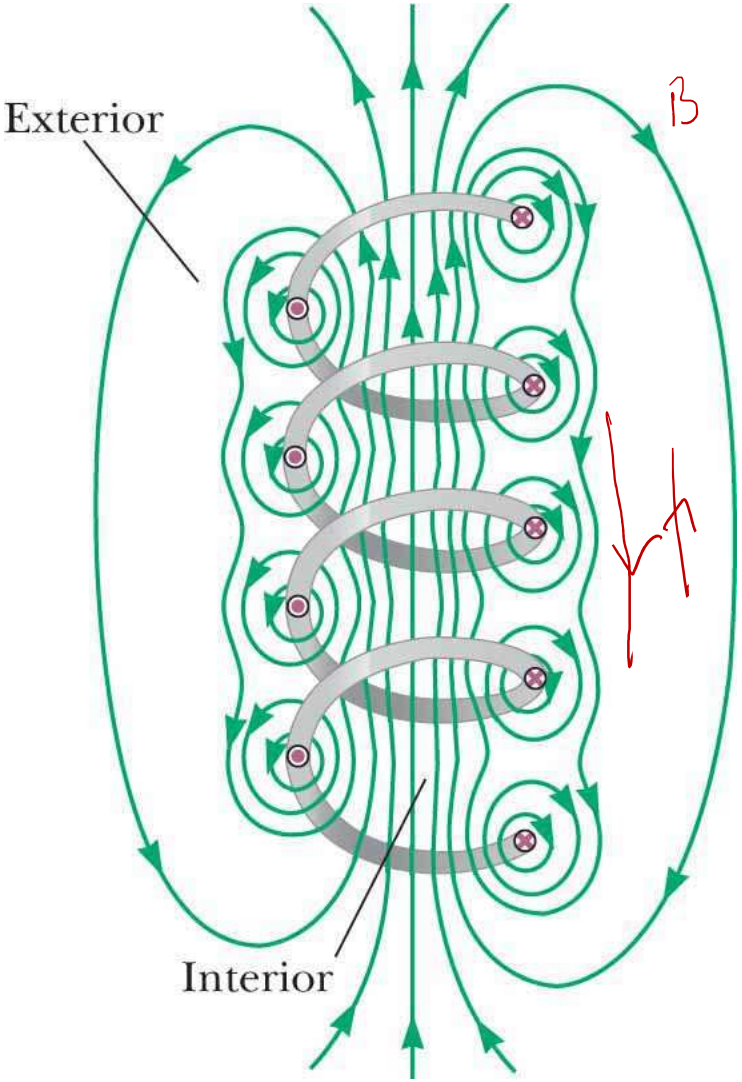
Example 29.6: The Magnetic Field Created by a Toroid

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = B \oint ds = B(2\pi r) = \mu_0 NI$$

$$B = \frac{\mu_0 NI}{2\pi r}$$



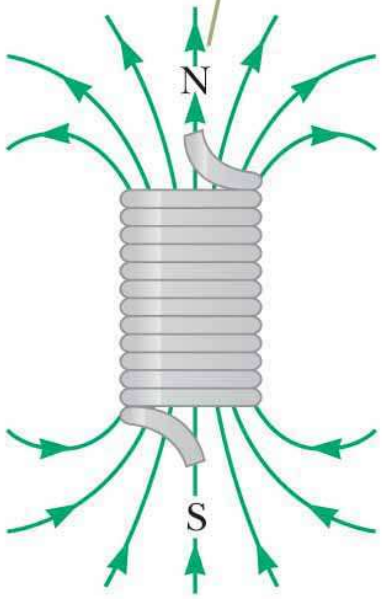
The Magnetic Field of a Solenoid



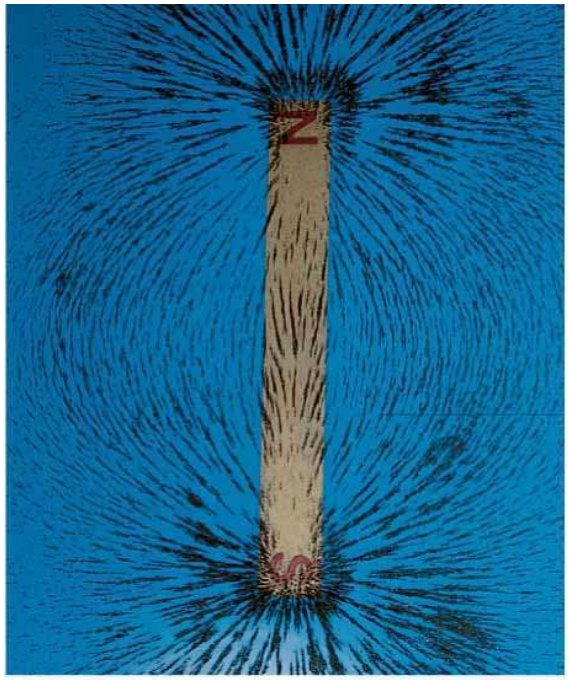
$$\oint \vec{B} \cdot d\vec{s} = \mu_0 I_{enc}$$

The Magnetic Field of a Solenoid

The magnetic field lines resemble those of a bar magnet, meaning that the solenoid effectively has north and south poles.



a



b

The Magnetic Field of a Solenoid

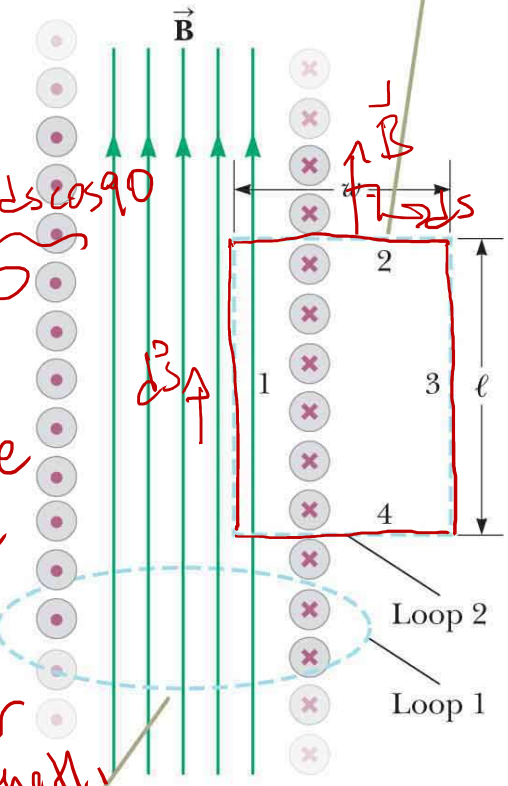
$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = \oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} + \oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = B\ell$$

path 1 path 2

$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{s}} = B\ell = \mu_0 NI$$

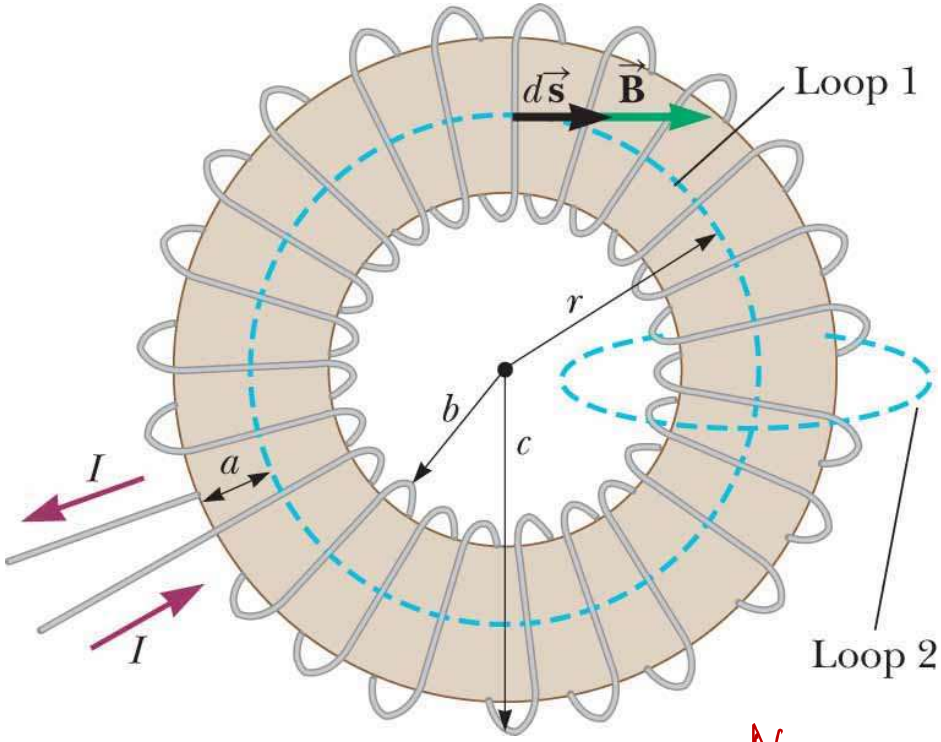
$$B = \mu_0 \frac{N}{\ell} I = \mu_0 nI$$

Ampère's law applied to the rectangular dashed path can be used to calculate the magnitude of the interior field.



Ampère's law applied to the circular path whose plane is perpendicular to the page can be used to show that there is a weak field outside the solenoid.

The Magnetic Field of a Solenoid



$$B_{\text{toroid}} = \frac{\mu_0 NI}{2\pi r} \qquad B_{\text{solenoid}} = \mu_0 nI \Rightarrow n = \frac{N}{2\pi r}$$

Quick Quiz 29.5

Consider a solenoid that is very long compared with its radius. Of the following choices, what is the most effective way to increase the magnetic field in the interior of the solenoid?

(a) double its length, keeping the number of turns per unit length constant

(b) reduce its radius by half, keeping the number of turns per unit length constant

(c) overwrap the entire solenoid with an additional layer of current-carrying wire

Quick Quiz 29.5

Consider a solenoid that is very long compared with its radius. Of the following choices, what is the most effective way to increase the magnetic field in the interior of the solenoid?

(a) double its length, keeping the number of turns per unit length constant

(b) reduce its radius by half, keeping the number of turns per unit length constant

(c) overwrap the entire solenoid with an additional layer of current-carrying wire

Gauss's Law in Magnetism

Magnetic Flux

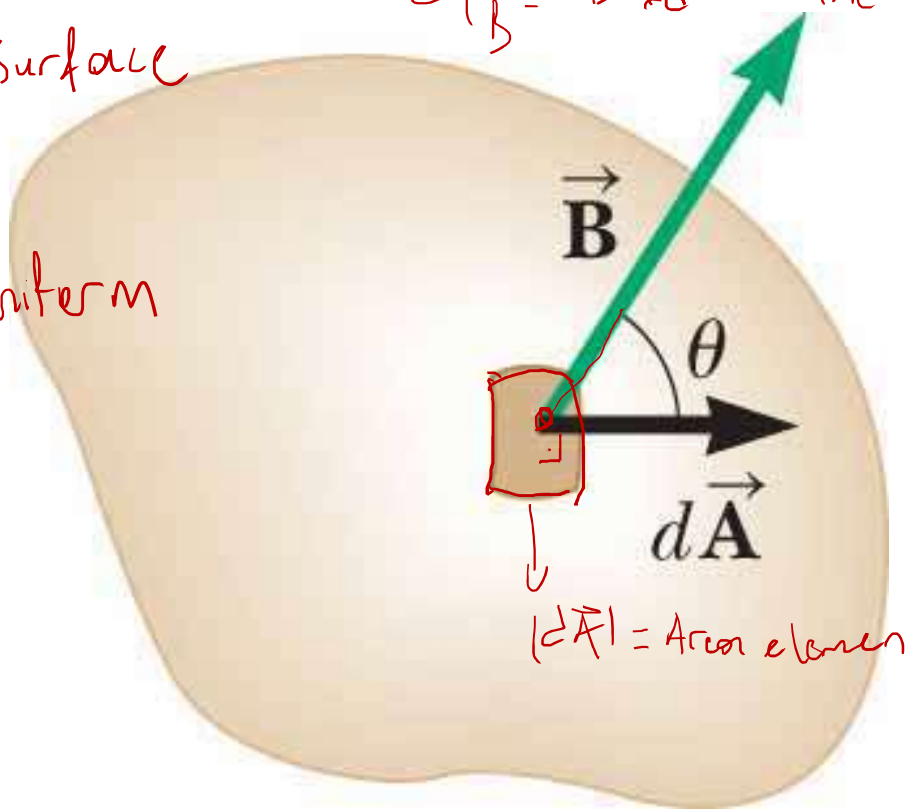
$d\Phi_B = \vec{B} \cdot d\vec{A} = |\vec{B}||d\vec{A}|\cos\theta$

Total flux through the surface

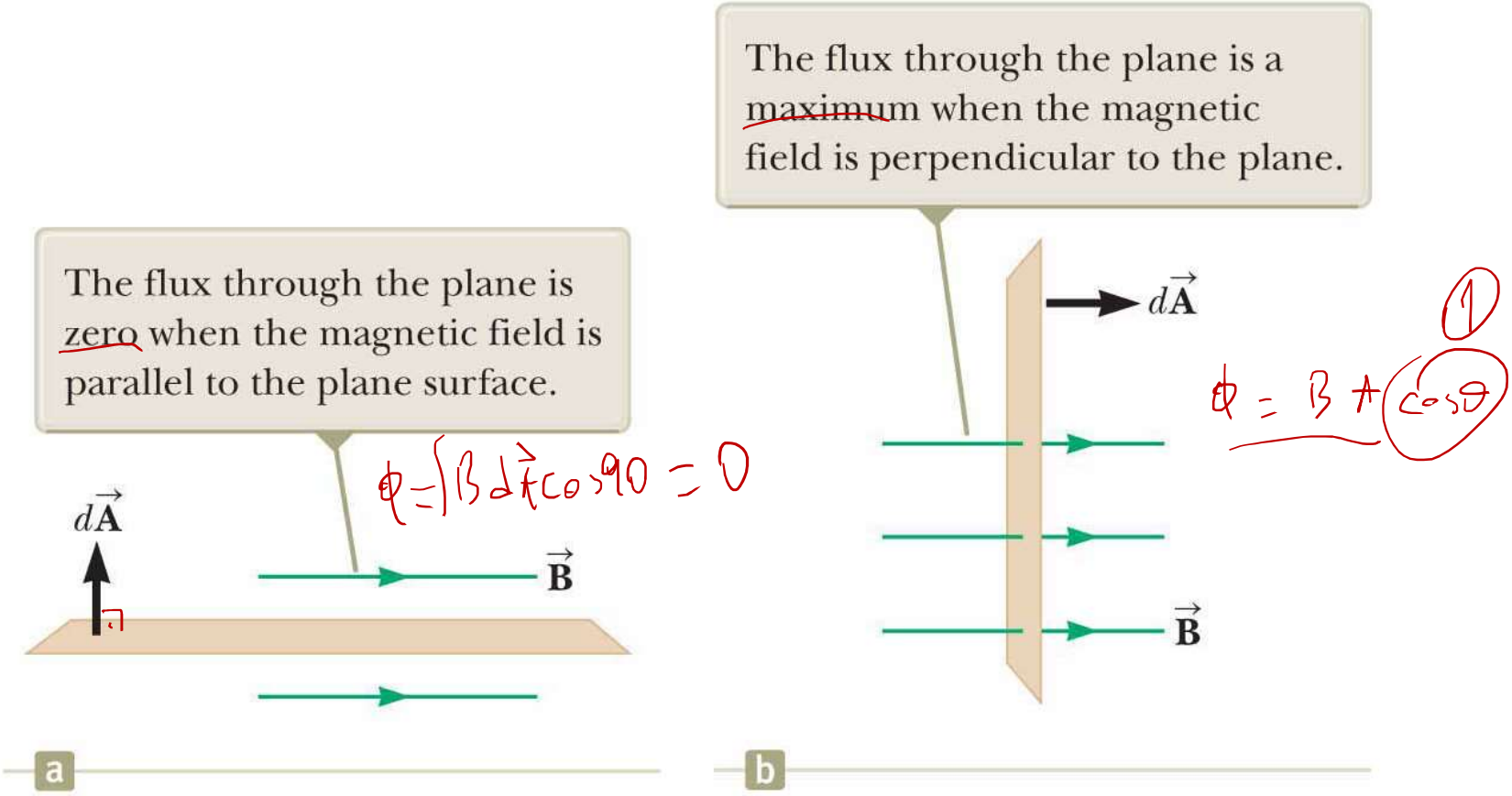
$$\Phi_B \equiv \int \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}}$$

If the magnetic field uniform

$$\Phi_B = BA\cos\theta$$



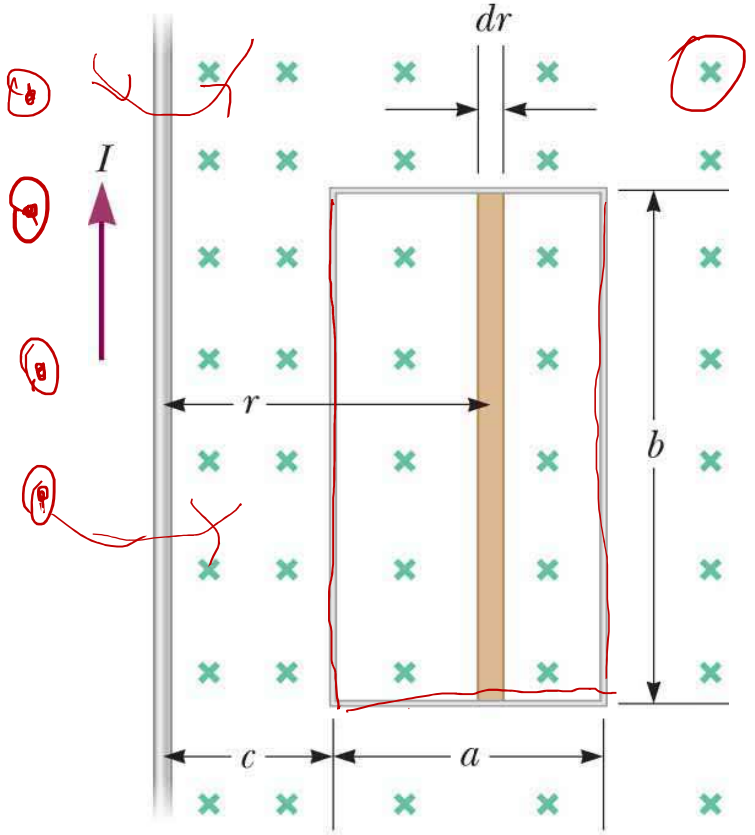
Gauss's Law in Magnetism



$$1\ \text{Wb} = 1\ \text{T} \cdot \text{m}^2$$

Example 29.7: Magnetic Flux Through a Rectangular Loop

A rectangular loop of width a and length b is located near a long wire carrying a current *I*. The distance between the wire and the closest side of the loop is *c*. The wire is parallel to the long side of the loop. Find the total magnetic flux through the loop due to the current in the wire.



Example 29.7: Magnetic Flux Through a Rectangular Loop

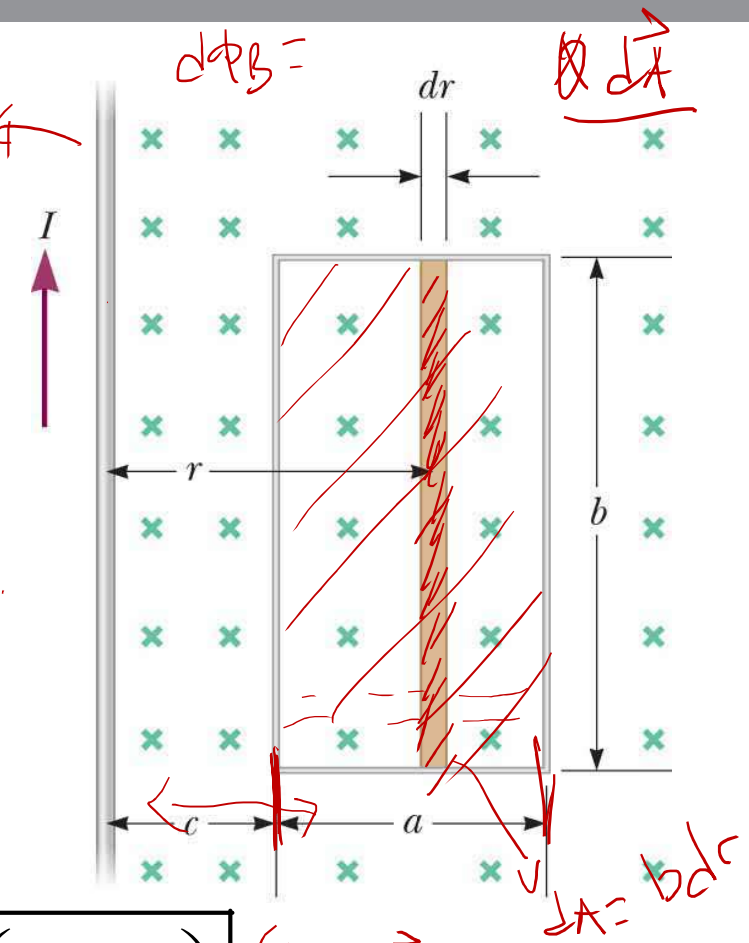
$$\Phi_B = \int \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = \int B\, dA$$

$$= \int \frac{\mu_0 I}{2\pi r} dA$$

$$\Phi_B = \int \frac{\mu_0 I}{2\pi r} b\, dr = \frac{\mu_0 I b}{2\pi} \int \frac{dr}{r}$$

$$\Phi_B = \frac{\mu_0 I b}{2\pi} \int_c^{a+c} \frac{dr}{r} = \frac{\mu_0 I b}{2\pi} \ln r \Big|_c^{a+c}$$

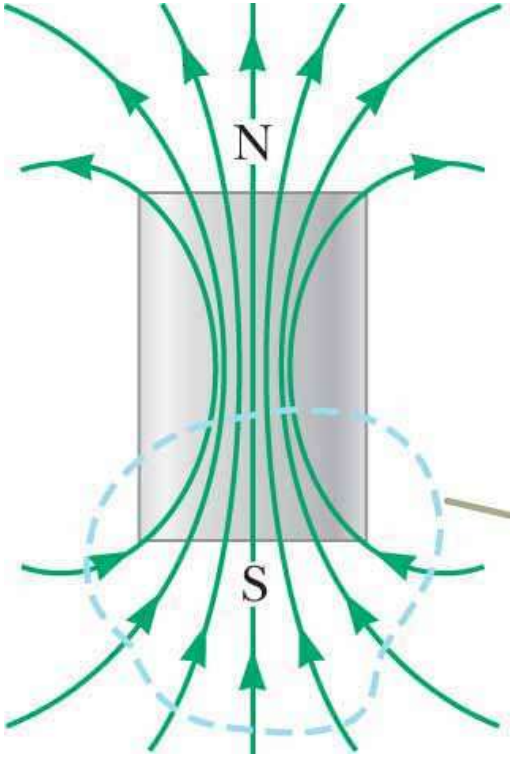
$$= \frac{\mu_0 I b}{2\pi} \ln\left(\frac{a+c}{c}\right) = \boxed{\frac{\mu_0 I b}{2\pi} \ln\left(1+\frac{a}{c}\right)}$$



Gauss's Law in Magnetism

The net magnetic flux through any closed surface is always zero:

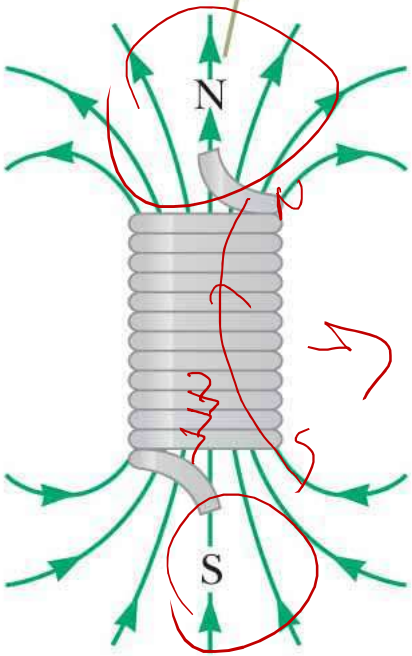
$$\oint \vec{\mathbf{B}} \cdot d\vec{\mathbf{A}} = 0$$



The net magnetic flux through a closed surface surrounding one of the poles or any other closed surface is zero.

Magnetism in Matter

The magnetic field lines resemble those of a bar magnet, meaning that the solenoid effectively has north and south poles.



a

current loop has a magnetic dipole moment

The Magnetic Moments of Atoms

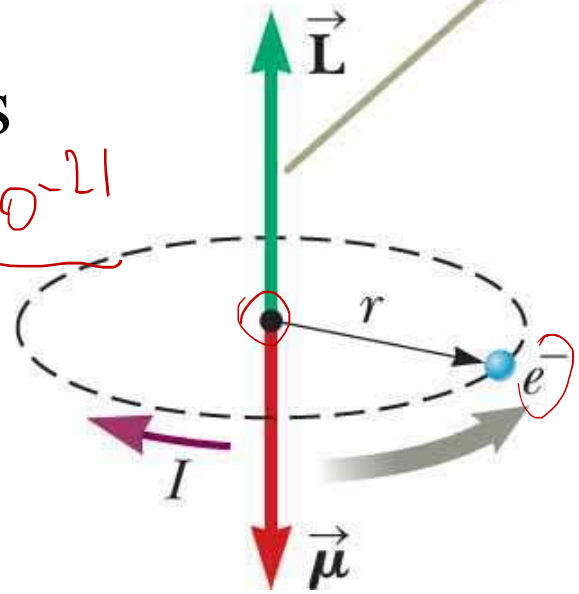
$$\mu = IA$$

$$L = m_e v r$$

$$\hbar = \frac{h}{2\pi} = 1.05 \times 10^{-34}\ \mathrm{J \cdot s}$$

$$\mu = \sqrt{2}\,\frac{e}{2m_e}\hbar$$

The electron has an angular momentum $\vec{\mathbf{L}}$ in one direction and a magnetic moment $\vec{\boldsymbol{\mu}}$ in the opposite direction.



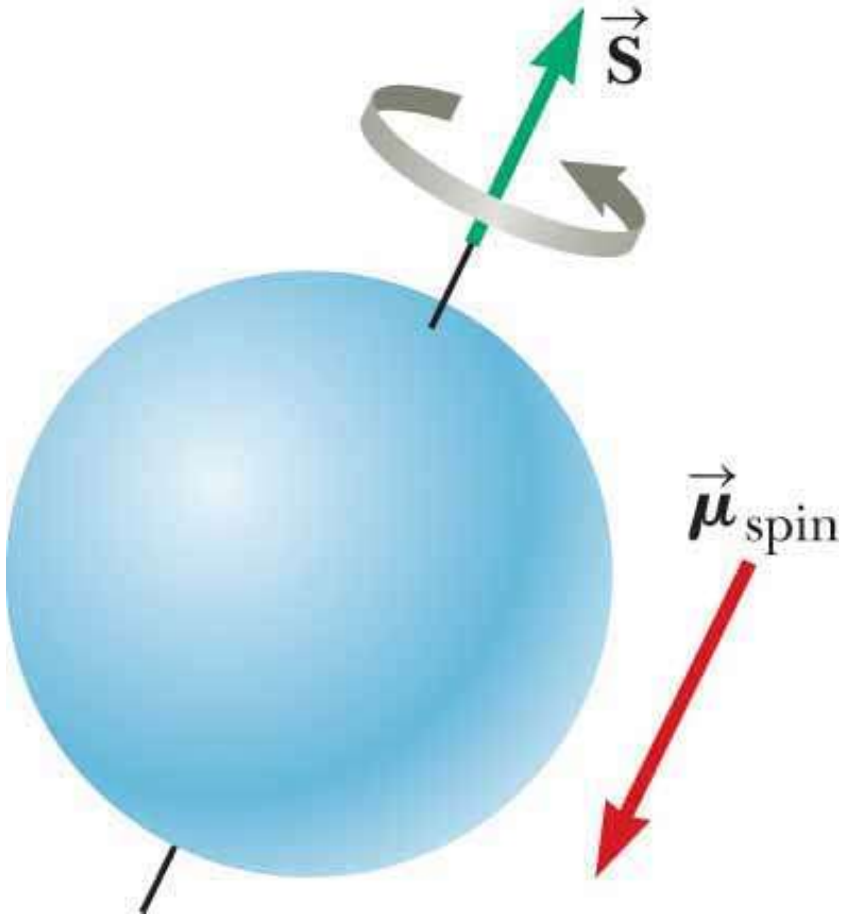
The Magnetic Moments of Atoms

Spin angular momentum

$$S = \frac{\sqrt{3}}{2}\hbar$$

$$\mu_{\text{spin}} = \frac{e\hbar}{2m_e}$$

$$\mu_{\text{B}} = \frac{e\hbar}{2m_e} = 9.27\times10^{-24}\ \text{J/T}$$



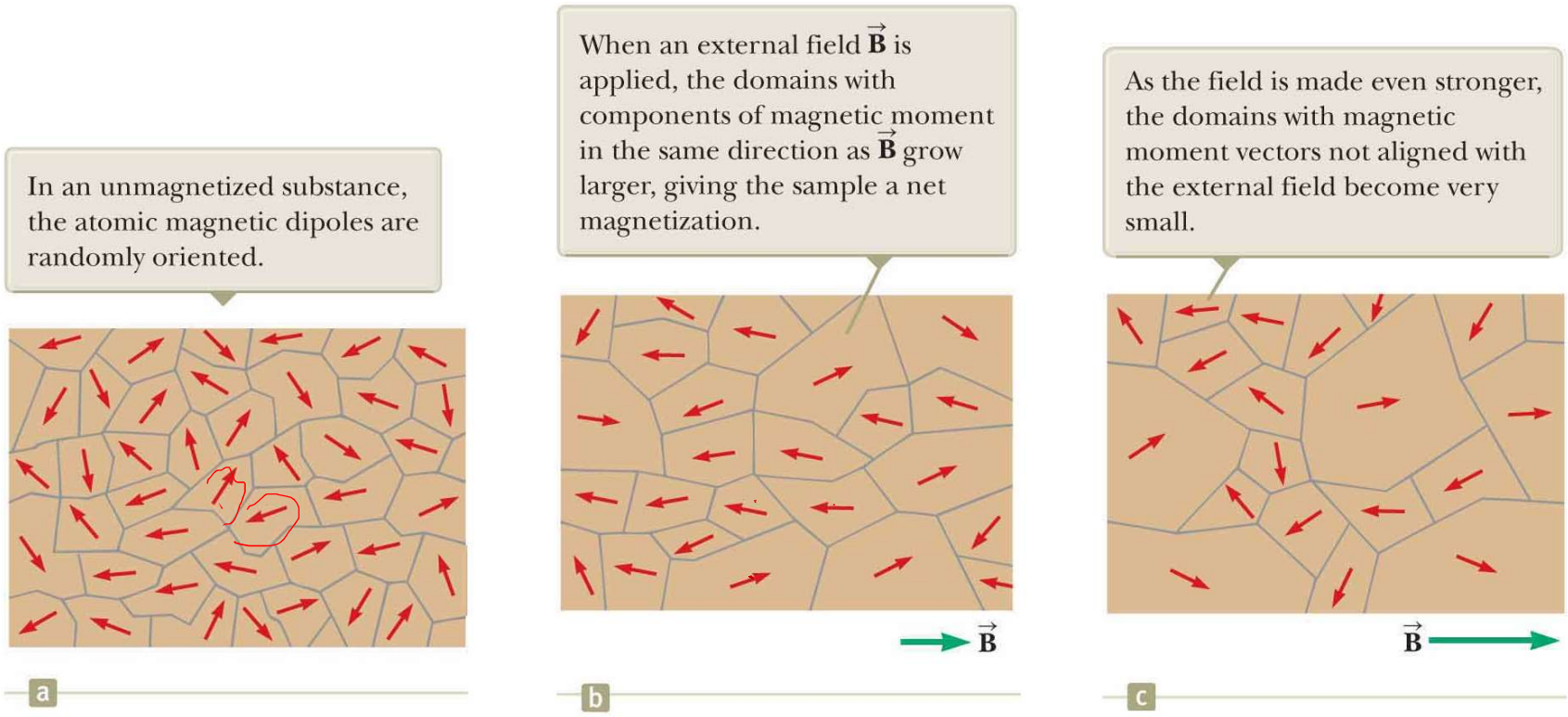
The Magnetic Moments of Atoms

TABLE 29.1 Magnetic Moments of Some Atoms and Ions

Atom or Ion	Magnetic Moment (10^{-24} J/T)
H	9.27
He	0
Ne	0
Ce^{3+}	19.8
Yb^{3+}	37.1

$$\mu_B = \frac{e\hbar}{2m_e} = 9.27 \times 10^{-24} \text{ J/T}$$

Ferromagnetism

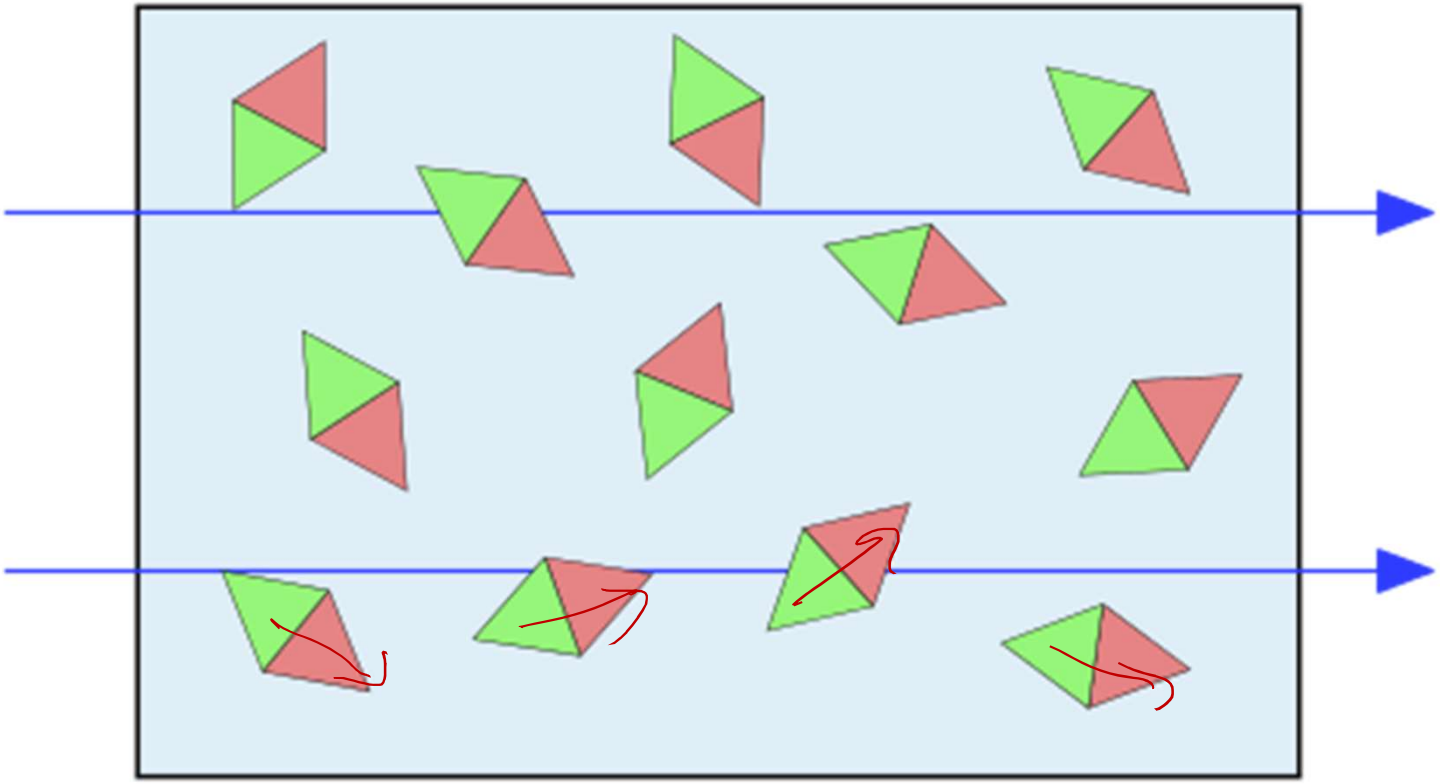


Ferromagnetism

TABLE 29.2 Curie Temperatures for Several Ferromagnetic Substances

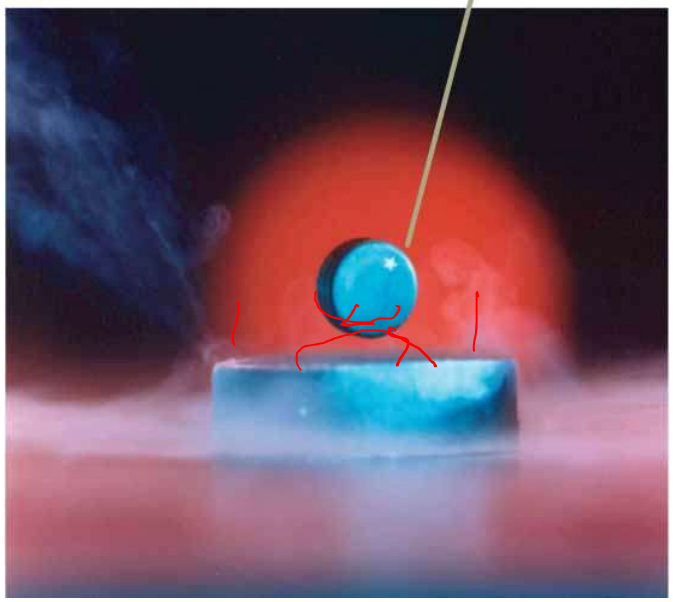
Substance	T_{Curie} (K)
Iron	1 043
Cobalt	1 394
Nickel	631
Gadolinium	317
Fe_2O_3	893

Paramagnetism



Diamagnetism

In the Meissner effect, the small magnet at the top induces currents in the superconducting disk below, which is cooled to −321°F (77 K). The currents create a repulsive magnetic force on the magnet causing it to levitate above the superconducting disk.



Assessing to Learn

Consider a long thin straight wire with a current I.
Which of the following statements about the magnetic field lines is true?

A. Field lines are parallel to the wire.
B. Field lines are perpendicular to the wire.
C. Field lines are directed radially away from the wire.
D. Field lines are circles centered on any point on the wire.



1. A only
2. B only
3. C only
4. D only
5. A and C only
6. B and D only
7. B and C only
8. None of them is true.

Assessing to Learn

A very long wire lies in a plane with a short wire segment. The long wire carries current I, while the short wire of length L carries current i. The two wires are parallel to each other. Which of the following statements are true?

A. The direction of the magnetic force exerted by the long wire on the short wire is directed away from the long wire.

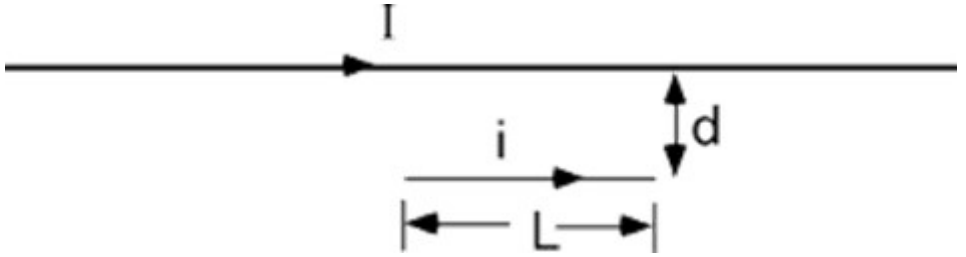
B. The magnitude of the force on the short wire is $\mu_0 IiL/2\pi d$.

C. The long wire experiences a force of exactly the same magnitude as the force experienced by the short wire.

1. A only 2. B only 3. C only 4. A and B

5. A and C 6. B and C 7. A, B and C

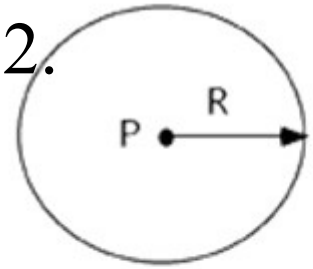
8. None of them is true.

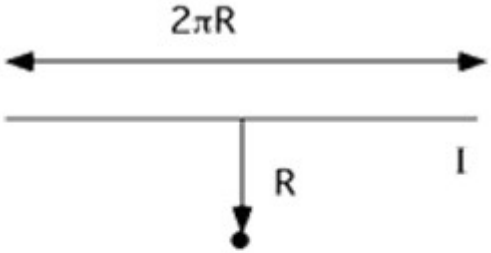


Assessing to Learn

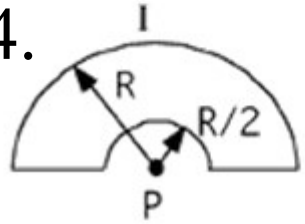
In all cases the wire shown carries a current *I*. For which situation is the magnitude of the magnetic field maximum at the point P?

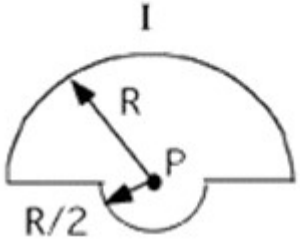
1. 2.





3. 4.

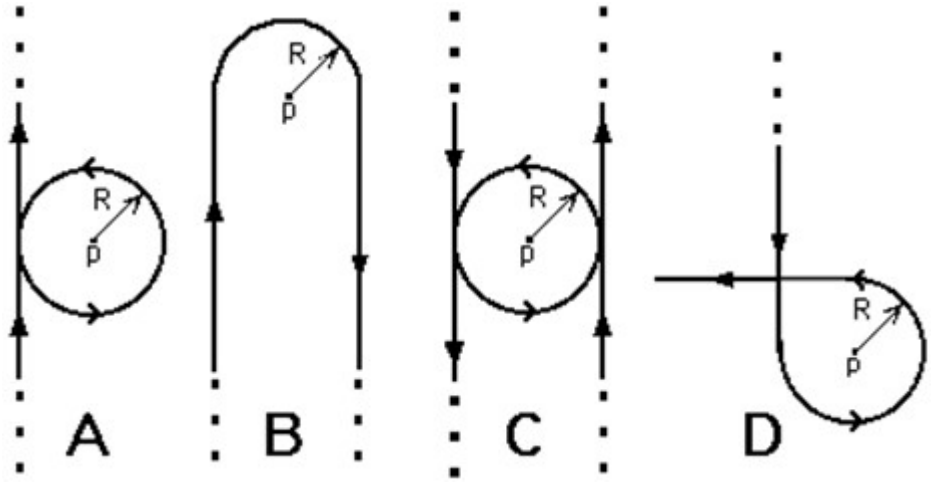




Assessing to Learn

Order the following situations according to the magnitude of the magnetic field at the point P. Order from highest to lowest.

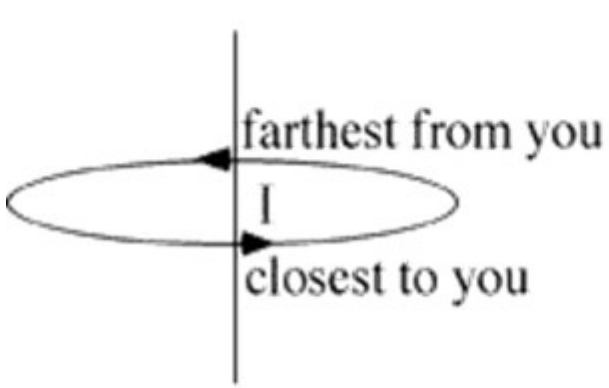
1. ABCD
2. ADBC
3. BDAC
4. CADB
5. DABC
6. None of the above

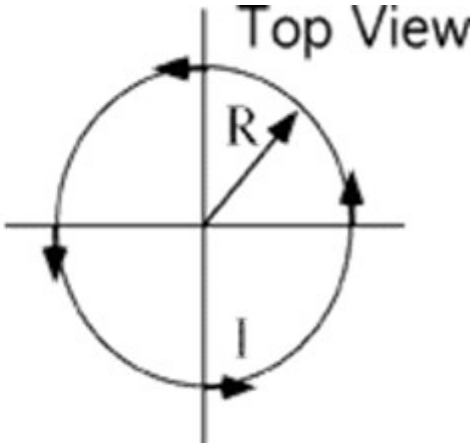


Assessing to Learn

The diagram shows a circular wire loop of radius *R* carrying current I. What is the direction of the magnetic field, B, at the center of the loop?

1. Left
2. Right
3. Up
4. Down
5. None of the above





Assessing to Learn

The diagram shows a circular wire loop of radius *R* carrying current I. What is the magnitude of the magnetic field, B, at the center of the loop?

1. 0
2. $\mu_0 I/4\pi R$
3. $\mu_0 I/2\pi R$
4. $\mu_0 I/4R$
5. $\mu_0 I/2R$
6. None of the above.

